

Positioning in Global Value Chains: World Map and Indicators, a New Dataset Available for GVC Analyses

M. Mancini, P. Montalbano, S. Nenci , and D. Vurchio

Abstract

This work reviews and computes the commonly used Global Value Chains (GVC) positioning indicators found in the empirical literature, providing scholars with a novel and comprehensive global dataset of upstreamness and downstreamness measures. This dataset covers a wide range of countries, including many developing nations, and industries, and spans an extensive timeframe. Specifically, it offers GVC positioning indicators for all economies and industries included in prominent Inter-Country Input-Output tables, such as ADB, EORA, OECD TiVA, WIOD, and Long-run WIOD. This work also delves into the degree of comparability across the different datasets, offering informative comparisons of the GVC positioning measures encompassing overlapping countries and periods, sectors, geographical regions, and income levels. Notably, these indicators are “ready-to-use” and open access, presenting an exceptional opportunity for qualitative and quantitative analyses of various economic dimensions on GVCs and for informing policymaking.

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1. Introduction

Over the last decades, the world economy has experienced a radical transformation through a significant fragmentation in the production of goods and services, and a deeper international division of labor. This

Michele Mancini is a Research Economist at the Bank of Italy, Italy; his email address is Michele.Mancini@bancaditalia.it. Pierluigi Montalbano is Full Professor of International Economic Policy at the Department of Social Sciences and Economics, Sapienza University of Rome, Italy, and Associate Faculty at the Department of Economics, Sussex University, Brighton, UK; his email is pierluigi.montalbano@uniroma1.it. Silvia Nenci (corresponding author) is Associate Professor of Economic Policy at the Department of Economics, Roma Tre University, Rome, Italy; her email address is silvia.nenci@uniroma3.it. Davide Vurchio is Assistant Professor of Economics in the Department of Economics and Finance, University of Bari Aldo Moro, Bari, Italy; his email address is davide.vurchio@uniba.it. The authors thank the editor and two anonymous referees for their valuable comments and suggestions, which have significantly enhanced the quality of this work. The authors also extend their appreciation to all the participants of the following workshops and conferences for their insightful comments and recommendations: 22nd ETSG Annual Conference, Ghent University, Belgium, 9–11 September 2021; VI International Workshop on GVCs “European Trade Policy and Global Value Chains,” Roma Tre University, Rome, 10 June 2022; 63rd Annual Conference of the Italian Economic Association, University of Turin, 20–22 October 2022; ITSG Conference, University of Florence, 1–2 December 2022. The views expressed in this work are purely those of the authors and may not in any circumstances be regarded as stating an official position of their respective institutions. A supplementary online appendix is available with this article at *The World Bank Economic Review* website.

transformation, which has been called the “Age of Global Value Chains” (World Bank 2020; Antràs and Chor 2022), has had remarkable effects worldwide via changes in income, productivity, and poverty (World Bank 2020).

Scholars have developed various indicators to map and measure the degree of participation of countries and sectors in GVCs (see, among others, Hummels, Ishii, and Yi 2001; Johnson and Noguera 2012; Koopman, Wang, and Wei 2014; Borin and Mancini 2023). A parallel body of work looks instead at the country and/or sectoral positioning *within* GVC, that is whether a given country (or industry) is specialized in relatively upstream activities or activities that are more proximate to final demand (see Antràs et al. 2012; Fally 2012; Antràs and Chor 2013; Miller and Temurshoev 2017; Wang et al. 2017; Fally and Hillberry 2018; Alfaro et al. 2019).¹ Such notions of production staging are important as they can contribute to model productivity differences or geography/firm organizational decisions, demonstrating that these choices are not merely a function of marginal costs (see Antràs and Chor 2022 for details). Together with the GVCs’ participation indicators, these positioning measures enrich and complete empirical analyses on GVCs and inform policymaking.

This paper aims to review and compute the most common GVC positioning indicators used in the empirical literature and provides scholars with a new global dataset of upstreamness and downstreamness measures for the largest number of countries—including most developing countries—and sectors and for the longest time span. Specifically, it provides GVC positioning indicators for all the economies and industries available in the most used Inter-Country Input Output (ICIO) tables, namely ADB MRIO (ADB kidb.adb.org/mrio); EORA (Lenzen et al. 2013); OECD TiVA (oe.cd/tiva); WIOD (Timmer et al. 2015); Long-run WIOD (Woltjer, Gouma, and Timmer 2021). This vast and comprehensive dataset—which will be regularly updated—is available at the following link <https://www.tradeeconomics.com/position/>.² It complements the already available database on GVC participation measures by the World Bank WITS (Borin, Mancini, and Taglioni 2021). Using these new measures, scholars can investigate the evolution of GVC positioning over time, at both the country and industry levels, and provide an international comparison. The key added value of this work is that scholars working on the topic of GVC and belonging to different research disciplines—economics, sociology, international economics, economic geography, international political economy, supply chain management, and international business—will benefit from these ready-to-use indicators, without necessarily getting into technicalities and performing matrix calculations.

Specifically, two measures are computed: (1) a measure of distance or *upstreamness* of a production sector from final demand, which was developed by Antràs et al. (2012), Fally (2012), and Antràs and Chor (2013, 2019); this measure captures the average number of production stages by pegging the endpoint of the sequence at final consumption, which makes it possible to measure the distance to the final demand of a product (or a country) along the production chains; and (2) a measure of distance or *downstreamness* of a given sector (or a country) from the economy’s primary factors of production (or sources of value-added), originally proposed by Fally (2012). This measure is based on a country-industry pair’s use of intermediate inputs and primary factors of production. To the best of the authors’ knowledge, this is the first work that computes GVC positioning indicators for such a large set of countries and sectors, and makes them freely available to scholars.

This paper illustrates the indicators included in this new open-access dataset, the methodologies applied to compute them, as well as the diverse methodologies and assumptions employed in constructing the various ICIO sources, offering valuable insights into their suitability based on users’ research interests. It also delves into the actual degree of comparability across the different datasets, offering informative

1 This definition is sufficiently broad to encompass various possible structures and distinctive characteristics of different GVCs (Antràs and Chor 2022).

2 Codes to replicate the dataset, as well as to compute positioning measures on other ICIO tables, are available here https://www.tradeeconomics.com/position/docs/position_codes.zip

comparisons of the GVC positioning metrics derived from these datasets. These comparisons encompass overlapping countries and periods, sectors, geographical regions, and income levels. Overall, while acknowledging nuanced differences, a commendable level of consistency among the various GVC positioning measures across the ICIO datasets has been identified.

The paper is structured as follows: Section 2 outlines the methodology employed for constructing the GVC indicators. Section 3 provides an overview of the ICIO data sources used in the computation of the GVC positioning measures. Section 4 offers descriptive statistics and explores the comparability of the indicators across different ICIO datasets. Section 5 concludes.

2. Measuring GVCs Positioning: The Upstreamness and Downstreamness Indicators

To compute the upstreamness or downstreamness of specific industries and countries, a common approach is to consider the extent to which a country-industry pair sells its output for final use to consumers worldwide or instead sells intermediate inputs to other producing sectors in the world. A sector that sells disproportionately to final consumers would appear to be downstream in value chains. In contrast, a sector that sells little to final consumers is more likely to be upstream in value chains.

Following this approach, two measures of GVC positioning, that are the most popular in the literature, have been computed in this work. The first indicator is a measure of distance or upstreamness of a production sector from final demand, which was developed by [Antràs et al. \(2012\)](#), [Fally \(2012\)](#), and [Antràs and Chor \(2013\)](#).³ Fally's model, as well as the variation proposed by [Antràs et al. \(2012\)](#), captures the average number of production stages by pegging the endpoint of the sequence at final consumption, which makes it possible to measure the *distance to the final demand* of a sector along the production chains. More specifically, this measure (labeled U in [Antràs and Chor 2019](#) and given the same name in the present dataset) aggregates information on the extent to which "an industry in a given country produces goods that are sold directly to final consumers or that are sold to other sectors that themselves sell disproportionately to final consumers. A relatively upstream sector is thus one that sells a small share of its output to final consumers and instead sells disproportionately to other sectors that themselves sell relatively little to final consumers" ([Antràs and Chor 2019](#), p. 127). Expanding upon these concepts, final goods are positioned one step away from immediate demand, inputs directly used in the production of final goods stand two steps away from demand, and inputs used in the production of other inputs reside three steps removed from immediate demand, and so forth. Furthermore, this count is weighted by the share of the value of output at each production stage in total output. The U indicator can assume values equal to or greater than 1: larger values are associated with relatively higher levels of upstreamness of the output originating from one sector.

The second measure, originally proposed by [Fally \(2012\)](#), is based on a country-industry pair's use of intermediate inputs and primary factors of production. It captures the distance or downstreamness of a given sector from the economy's primary factors of production (or sources of value-added). According to this measure (labeled D), an industry in each country is downstream if its production process embodies a larger value of intermediate inputs relative to its use of primary factors of production. Conversely, if an industry relies disproportionately on value-added from primary factors of production, then this industry is relatively less downstream. The D indicator can assume values equal to or greater than 1: larger values are associated with relatively higher levels of downstreamness of an industry.

These positioning indicators are computed by using the intermediate use matrix (Z), the final demand matrix (FD), and the value-added matrix (VA). Following [Antràs et al. \(2012\)](#) and [Antràs and Chor \(2019\)](#), a "net inventory" correction has been performed. This correction consists of imputing N_i^* changes

3 Though the arguments used to develop the index differ in [Fally \(2012\)](#) and [Antràs et al. \(2012\)](#), [Antràs and Chor \(2013\)](#) emphasize that the resulting indexes are equivalent.

in inventories in country i , sector r , to each Z_{ij}^{rs} intermediates sold by country i sector r to country j sector s , and FD_{ij}^r final goods in sector r sold by i to j , by applying a multiplicative factor equal to $Y_i^r/(Y_i^r - N_i^r)$ where Y_i^r is the gross output in sector r in country i and is computed as follows:⁴

$$Y_i^r = \sum_{s=1}^S \sum_{j=1}^J Z_{ij}^{rs} + \sum_{j=1}^J FD_{ij}^r \quad (1)$$

In order to measure sectoral upstreamness, the U_i^r index by [Antràs and Chor \(2013\)](#) has been adopted. Since $Y_i^r = \sum_{s=1}^S \sum_{j=1}^J a_{ij}^{rs} Y_j^s + FD_{ij}^r$ (where $a_{ij}^{rs} = Z_{ij}^{rs}/Y_j^s$ is the dollar amount of sectors r 's output from country i needed to produce one dollar worth of industry s 's output in country j), by iterating such identity, industry r 's output in country i can be expressed as an infinite sequence of terms as follows:

$$Y_i^r = FD_{ij}^r + \sum_{s=1}^S \sum_{j=1}^J a_{ij}^{rs} FD_j^s + \sum_{s=1}^S \sum_{j=1}^J \sum_{t=1}^S \sum_{k=1}^J a_{ij}^{rs} a_{jk}^{st} FD_k^t + \dots \quad (2)$$

As in [Antràs and Chor \(2019\)](#), the weighted average position of each country/sector pair has been computed by multiplying each term by its respective production-staging distance from final use plus 1 and dividing everything by Y_i^r . This means that the first term in equation (2), representing the production stage destined to final consumption, is multiplied by 1, the second term in equation (2), representing the production stage one step before the completion of final good, is multiplied by 2, and so on. Building on such identity, the upstreamness index can be expressed as follow:

$$U_i^r = 1 * \frac{FD_{ij}^r}{Y_i^r} + 2 * \frac{\sum_{s=1}^S \sum_{j=1}^J a_{ij}^{rs} FD_j^s}{Y_i^r} + 3 * \frac{\sum_{s=1}^S \sum_{j=1}^J \sum_{t=1}^S \sum_{k=1}^J a_{ij}^{rs} a_{jk}^{st} FD_k^t}{Y_i^r} + \dots \quad (3)$$

It can be shown that in matrix notation this corresponds to:

$$U = [I - A]^{-2} FD_i^r \oslash [I - A]^{-1} FD_i^r \quad (4)$$

Where A is $J \times S$ -by- $J \times S$ matrix of the a_{ij}^{rs} s whereas $\oslash$ refers to an elementwise division.

As for downstreamness, following [Antràs and Chor \(2013\)](#), the D_j^s measure has been adopted. Since $Y_j^s = \sum_{r=1}^S \sum_{i=1}^J Z_{ij}^{rs} + VA_j^s = \sum_{r=1}^S \sum_{i=1}^J b_{ij}^{rs} Y_i^r + VA_j^s$ (where $b_{ij}^{rs} = Z_{ij}^{rs}/Y_i^r$ is the share of sectors r 's output in country i that is used in industry s in country j), and $Y_j^s = VA_j^s + \sum_{r=1}^S \sum_{i=1}^J b_{ij}^{rs} VA_i^r + \sum_{r=1}^S \sum_{i=1}^J \sum_{t=1}^S \sum_{k=1}^J b_{ki}^{tr} b_{ij}^{rs} VA_k^t + \dots$, analogously to the upstreamness case, downstreamness is computed as follows:

$$D_j^s = 1 * \frac{VA_j^s}{Y_j^s} + 2 * \frac{\sum_{r=1}^S \sum_{i=1}^J b_{ij}^{rs} VA_i^r}{Y_j^s} + 3 * \frac{\sum_{r=1}^S \sum_{i=1}^J \sum_{t=1}^S \sum_{k=1}^J b_{ki}^{tr} b_{ij}^{rs} VA_k^t}{Y_j^s} + \dots \quad (5)$$

In equation (4) each element is multiplied by the production stage distance from primary factors plus 1 and divided by country s gross output in sector j . In matrix notation:

$$D = [I - B]^{-2} VA_j^s \oslash [I - B]^{-1} VA_j^s \quad (6)$$

Where B is $J \times S$ -by- $J \times S$ matrix of the b_{ij}^{rs} whereas $\oslash$ refers to an elementwise division.

4 Value-added is obtained as $VA_j^s = Y_j^s - \sum_{r=1}^S \sum_{i=1}^J Z_{ij}^{rs}$. In doing so, the analysis disregards taxes less subsidies on products, CIF/FOB adjustments on exports, direct purchases abroad by residents, purchases on the domestic territory by nonresidents, and international transport margins. Therefore, the different methods used by data producers to compute these elements do not affect the results. This is a standard procedure in the literature, see [Antràs and Chor \(2019\)](#) and [Caliendo and Parro \(2015\)](#).

These measures were originally used to know where single industries are located along global value chains. Nevertheless, they can be aggregated at the country, global industry and global aggregate level. As a matter of fact, at the global aggregate level, upstreamness and downstreamness coincide and are a proxy for global production complexity (Antràs and Chor 2019).

3. Data Sources

ICIO tables offer an extensive overview of global transactions involving goods and services. Within a single integrated accounting framework, these tables combine the national input-output tables (which illustrate financial interactions between economic sectors within a country) of different economies at specific points in time, along with trade flow tables (depicting export and import values by country and economic sector). By encapsulating supply-use relationships between industries and spanning countries, these tables make it possible to delineate the vertical configuration of international production sharing. Moreover, they enable the quantification of cross-border value flows for a particular country or region (Inomata 2017).

Construction and Characteristics of the ICIO Datasets

Since the early 2000s, various research initiatives have undertaken the development of different ICIO databases. The construction of ICIO tables is a complex task that typically requires the application of specific compilation methods and assumptions to reconcile data from different sources and cope with data availability or reliability issues.⁵ As a result, divergent methodological choices can result in noteworthy differences among various databases (see Jones et al. 2014).⁶ For instance, the ways goods and services are categorized by end-use in the ICIO tables, as well as the inclusion of re-exports or processing trade, that notably affect economies engaged in such activities (e.g., the Netherlands and Belgium, as well as China and Mexico). Furthermore, deviations can arise from other factors, such as the origin of the raw data. The data used in creating ICIO tables are drawn from diverse statistical sources. These encompass national Input-Output tables, Supply-Use tables, national accounts, merchandise trade statistics categorized under the Broad Economic Categories classification or the Harmonized System, and trade in services statistics based on the balance of payments data, among others. Even if the ICIO tables were constructed using identical macroeconomic constraints, there remains considerable room for variation across the independent models at the sector level. The choice of aggregation or disaggregation level also represents an important factor contributing to variation. The target databases present different aggregation levels. This means that the national IO tables have to be reclassified, aggregated, or disaggregated, in order to adjust the source national tables to match the chosen sector classification. Additionally, discrepancies can stem from the update frequency and availability of data during the table's construction. ICIO tables are compiled at various points in time and might not undergo uniform revision or updates. These aspects also significantly influence the extent of country coverage and the temporal span covered in the ICIO tables.

Despite their increasing widespread use in economic research, ICIO tables share a number of shortcomings. First, because they rely on aggregated input-output data, the resulting sectoral disaggregation is rough. Therefore, data sources miss a notable amount of GVC-related trade activities taking place among sectors. Second, in constructing the tables, researchers are forced to impose strong assumptions to back

5 All ICIO databases incorporate a certain level of modeling. Within these databases, certain segments are characterized by being overdetermined, featuring multiple and occasionally conflicting reports. Conversely, other segments are underdetermined, necessitating the incorporation of assumptions or modeling techniques to complete sections of the dataset that lack coverage from official sources (Casella et al. 2019).

6 The topic of ICIO construction and reliability has been extensively discussed in the literature (see among others Wiedman et al. 2011; Dietzenbacher et al. 2013; Tukker and Dietzenbacher 2013; Owen et al. 2014; Tukker et al. 2020).

out some bilateral intermediate input trade flows that cannot be directly read from either customs data or national IO tables, leading to relevant aggregation biases (de Gortari 2019; Antràs 2020). For instance, the “proportionality assumption” which—due to the lack of data on the destination industries of international trade flows—implies identical trade shares for all input-purchasing industries (i.e., imported commodities are proportionally distributed over the target sectors) and the “production assumption,” which says that because of the aggregation level, each industry grouping produces all its different outputs using a single production function.⁷ The absence of information regarding heterogeneity in production processes within firm-level data is a notable gap in the prevailing databases. Specifically, the analysis of value chains could be enhanced by segmenting industries into various types of firms based on characteristics such as size, ownership, exporter status, or import share.⁸

Despite these limitations, ICIO tables have emerged as an essential resource for economists investigating GVCs. These tables serve as a means to quantify the involvement of countries and sectors in GVCs, as well as various aspects of GVC linkages (Antràs 2020; Antràs and Chor 2022).

ICIO Datasets Used for Calculating GVC Positioning Indicators

This study employs widely used and openly accessible databases to calculate the GVC positioning measures. Specifically, it makes use of the following ICIO datasets: EORA, OECD TiVA, WIOD, Long-run WIOD, and ADB (see table s1.1 in the supplementary online appendix, for a summary of the fundamental information).

The EORA Global Supply Chain Database (Lenzen et al. 2012; Lenzen et al. 2013) provides a set of both national and global input-output tables covering 189 countries from 1990 to 2022.⁹ More specifically, reference is made to EORA26, a simplified model where all countries have been aggregated to a common 26-sector harmonized classification (International Standard Industrial Classification of Economic Activities—ISIC—Rev. 3), and the supply-use tables from the full EORA Multi-Regional Input-Output (MRIO) have been converted to symmetric product-by-product IO tables. This dataset thus contains only symmetric product-by-product and industry-by-industry IO tables.¹⁰ The EORA data involve the integration of various data sources, including input-output tables and main aggregate data obtained from national statistical offices, as well as the UN National Accounts main aggregates database and international trade data. The database incorporates MRIO tables from national statistics offices of all generating countries. While EORA encompasses global coverage, it heavily relies on imputation techniques to address data gaps in countries with less developed statistical systems. For nations lacking official IO tables, estimates are formulated using a proxy input-output table. This proxy amalgamates macroeconomic data with a standardized input-output framework, based on an average derived from Australia, Japan, and the

- 7 More recent ICIO tables have sought to improve on this standard methodology, with a key step being to construct different proportionality weights for flows of imported intermediates and final goods respectively. The WIOD, the OECD-ICIO, as well as the most recent GTAP editions, have each implemented this approach.
- 8 In recent times, there have emerged some initiatives aimed at enhancing input-output tables and supply and use tables to include firm heterogeneity. Notably, the OECD-ICIO tables have been expanded to encompass firm heterogeneity by disaggregating countries, as demonstrated in the recently extended version available at <http://oe.cd/icio>. The Asian International Input-Output Tables by IDE-JETRO use confidential firm-level surveys from several Asian countries. In particular, these surveys provide insights into the breakdown of firm imports (Meng, Zhang, and Inomata 2013). The majority of these initiatives are conducted on a single-country basis, primarily driven by the requirement for substantial volumes of data (for details on these attempts, see Ahmad et al. 2023).
- 9 The present study has included data in its database for the years spanning 1990 to 2015, which are readily accessible through open-access sources. Regrettably, updates for more recent years are contingent upon licensing agreements and cannot be included in this dataset.
- 10 Please note that both the step of aggregating sectors from the higher sectoral detail of EORA to the lower detail of EORA26 and the step of converting Supply/Use tables to IO tables, involve a net information loss and the introduction of some new assumptions (see <https://worldmr.io/>).

United States. Consequently, EORA compilers stress that values should be interpreted as mean values with associated confidence intervals. In addition, these tables are underpinned by the standard ICIO assumptions (the “proportionality assumption” and the “production assumption”). Furthermore, the EORA tables prioritize the fulfillment of balancing conditions, primarily for larger economies. Consequently, these conditions are notably well satisfied (within a 1 percent imbalance) for major economies such as the United States, Japan, China, and Germany. The satisfaction of these conditions is less consistent for smaller countries or countries affected by periods of conflict, government transition, or hyperinflation. For more comprehensive information about the construction of the EORA database see <https://www.worldmrio.com>.

The 2022 edition of the OECD Inter-Country Input-Output (ICIO) Trade in Value Added (TiVA) database covers 76 economies (including all OECD, EU, G20, and ASEAN economies) and a selection of regional aggregates. Indicators are available for 45 industries within a hierarchy based on ISIC Rev. 4. The OECD’s ICIO system consists of a set of annual symmetric industry-by-industry global input-output tables. These are constructed using statistics compiled from national, regional, and international sources according to the 2008 System of National Accounts (2008 SNA). Specifically, harmonized national Supply-Use Tables (SUTs) and Input-Output tables are key inputs in the construction of the ICIO tables. The OECD TiVA uses the Broad Economic Categories (BEC) classification and end-use categories to distinguish imports, softening the “proportionality assumption” (i.e., in TiVA import shares by country might differ across use categories although they are equal within these categories). Over time, the OECD has engaged in various endeavors aimed at enhancing the scope and quality of the ICIO tables, in addition to producing annual tables.¹¹ For further details on OECD TiVA, see <https://www.oecd.org/sti/ind/measuring-trade-in-value-added.htm>.

The World Input-Output Database (WIOD) (Timmer et al. 2015), in its 2016 release, covers 43 countries for the period 2000–2014. This database classifies data for 56 sectors according to the ISIC Rev. 4. The tables adhere to the 2008 version of the SNA and are assembled utilizing a combination of published statistics from national statistical institutes across the globe, along with diverse international statistical sources. More specifically, WIOD uses SUTs from individual countries’ national accounts as the starting point to integrate with bilateral trade statistics and derive the final symmetric world I-O table (WIOT). This approach underpins a commendable level of data quality, albeit with the acknowledged constraint of limiting the number of countries covered. This trade-off between data quality and coverage is a central consideration. It is worth noting that, in contrast to the majority of datasets, it openly shares the methodologies employed in its creation. Distinguishing itself from other common datasets, the construction of WIOD employs an estimation approach for the import of goods that deviates from the conventional import “proportionality assumption” (Dietzenbacher et al. 2013).¹² For further details on WIOD, see <https://www.rug.nl/ggdc/valuechain/wiod>.

The Long-run WIOD (Woltjer, Gouma, and Timmer 2021) provides an extensive annual time-series of WIOTs spanning the years 1965 to 2000 (<https://www.rug.nl/ggdc/valuechain/long-run-wiod>). It includes 25 countries and 23 sectors (according to the ISIC Rev. 3.1.). This level of granularity in sector and country coverage reflects the common denominator across nations based on the available data. Just like WIOD, it is also an open-access database that offers complete details about construction methods, enabling others to

- 11 Given that only specific major economies (such as China, Japan, and the United States) consistently update and release SUTs, creating ICIO tables for more recent years involves extrapolating the latest harmonized SUTs. These extrapolations are guided by constraints derived from National Accounts time series data up to 2020. Subsequent enhancements are achieved through adjustments using balance-of-payments statistics and bilateral trade data for goods and services.
- 12 WIOD starts with imports as they are presented in the supply tables and subsequently employs bilateral trade statistics to compute import shares for three specific end-use categories: intermediate use, final consumption, and investment. This determination is guided by a refined interpretation of the well-known BEC codes that delineate end-use distinctions. It is essential to highlight that, even though within each of the three end-use categories the allocation continues to rely on a proportionality assumption, this occurs at a lower level of aggregation.

make use of the data. The foundation of the long-run WIOTs rests upon published national input-output tables, which have been collected from national statistical institutes. These tables were harmonized in terms of concepts and classifications to ensure consistency across countries. The interlinking of national input-output tables across countries is achieved through detailed bilateral international trade statistics. An underlying conceptual framework derived from the System of National Accounts 1993 (SNA 1993) forms the basis for constructing these tables. This approach stems from the broader availability of national accounts data conforming to SNA 1993 within the time range from 1965 to 2000, compared to SNA 2008, which typically lacks historical data coverage. Just like WIOD, bilateral trade statistics have been used to derive import shares for three end-use categories (intermediate use, final consumption use, or investment use). For more comprehensive insights into the Long-run WIOD, refer to [Woltjer, Gouma, and Timmer \(2021\)](#).

The ADB multiregion I-O database (ADB MRIO) has been developed by the Asian Development Bank. It is basically an extension of the WIOD to facilitate analysis work related to Asia and the Pacific Region. Nineteen Asian countries have been added into WIOD for the years 2000 and 2007 to 2022 (<https://kidb.adb.org/mrio>). The current ADB MRIO database encompasses 62 economies, with a focus on 25 Asian economies. Each of these economies is further categorized into 35 sectors based on ISIC Rev. 3.1. Notably, the data provided by this database stem from estimations produced by researchers, rather than being based solely on official statistics. These estimations are updated annually, incorporating the latest published statistics. Consequently, the ADB MRIO database stands out as one of the most up-to-date ICIO resources, boasting the broadest coverage of developing Asia.

As highlighted, discrepancies in the original construction of an ICIO database, necessitating modeling assumptions and computational steps, along with the consideration of specific data issues, can contribute to variations in the estimation of value-added trade data and GVC indicators. Given the unobservable nature of trade in value-added, assessing the precision of measures derived from various ICIO datasets presents a challenge. Consequently, it is important to note that ascribing inherent superiority to one database over another is unjustified. The choice between datasets hinges on specific research objectives. For instance, researchers seeking the broadest country coverage, particularly in the context of developing economies, may find the EORA database most suitable. On the other hand, those placing emphasis on data holding official status and a more refined sectoral breakdown could opt for WIOD. For those who value both data reliability and regular annual updates, the OECD TiVA database would be a suitable choice. Should their focus lie within a specific geographic region, such as Asia, the ADB MRIO database would be pertinent. If they need a historical perspective, they should certainly consider opting for the Long-run WIOD.

4. Descriptive Statistics, Evolution, and Comparability of GVC Positioning Measures Across ICIO Datasets

Although the construction of GVC positioning indicators across multiple ICIO sources naturally prompts questions concerning the comparability of these metrics and the relative positioning of countries over time, the inherent disparities in the characteristics and computation methods of the various ICIO datasets do not allow *naïve* comparisons of the computed GVC position indicators. Additionally, as already highlighted, the ICIO datasets not only differ in their computation methodologies but also in terms of their temporal, country, and sectoral coverage. Specifically, only 24 countries have data available in all the datasets, and the year 2000 is the sole year with comprehensive data across all datasets. In the other years, data availability varies: for instance, data for the 1995–99, 2001–2006, and 2015 are only available for three datasets, while in the most recent years (2016–2020), data is freely available only for two of them (OECD TiVA and ADB). ICIO datasets vary also in terms of sectoral coverage and classification ranging from the 26 (ISIC rev. 3) sectors of EORA to the 56 (ISIC rev. 4) sectors of WIOD.

Table 1. Comparisons between Inter-Country Input-Output Datasets and IMF World Economic Outlook

Variable	<i>p</i> 10	<i>p</i> 50	<i>p</i> 90	Mean	Variance	N
Trade % differences compared to IMF WEO						
IMF vs. ADB (exp)	18.37	41.79	90.39	51.18	2334.20	387
IMF vs. EORA (exp)	− 22.80	12.14	94.02	31.41	6302.54	387
IMF vs. TiVA (exp)	6.41	17.74	37.96	19.79	202.24	387
IMF vs. WIOD (exp)	21.49	49.74	100.95	60.55	3031.32	387
IMF vs. ADB (imp)	15.48	30.91	62.14	35.57	537.85	387
IMF vs. EORA (imp)	− 15.73	2.06	45.07	11.00	1366.55	387
IMF vs. TiVA (imp)	1.42	13.08	31.65	15.24	222.83	387
IMF vs. WIOD (imp)	17.55	35.14	83.12	44.28	965.69	387
Annual growth rates differences compared to IMF WEO						
IMF vs. ADB (exp)	− 3.61	0.72	5.87	1.24	34.74	301
IMF vs. EORA (exp)	− 6.61	− 0.08	7.18	0.34	56.34	387
IMF vs. TiVA (exp)	− 2.97	0.44	3.53	0.73	26.10	387
IMF vs. WIOD (exp)	− 4.16	0.80	6.60	1.11	42.14	344
IMF vs. ADB (imp)	− 2.99	0.46	4.80	1.05	26.98	301
IMF vs. EORA (imp)	− 6.17	− 0.21	6.22	0.32	56.37	387
IMF vs. TiVA (imp)	− 2.76	0.32	3.81	0.62	29.34	387
IMF vs. WIOD (imp)	− 3.14	0.52	4.57	1.06	26.22	344

Source: Authors' computation.

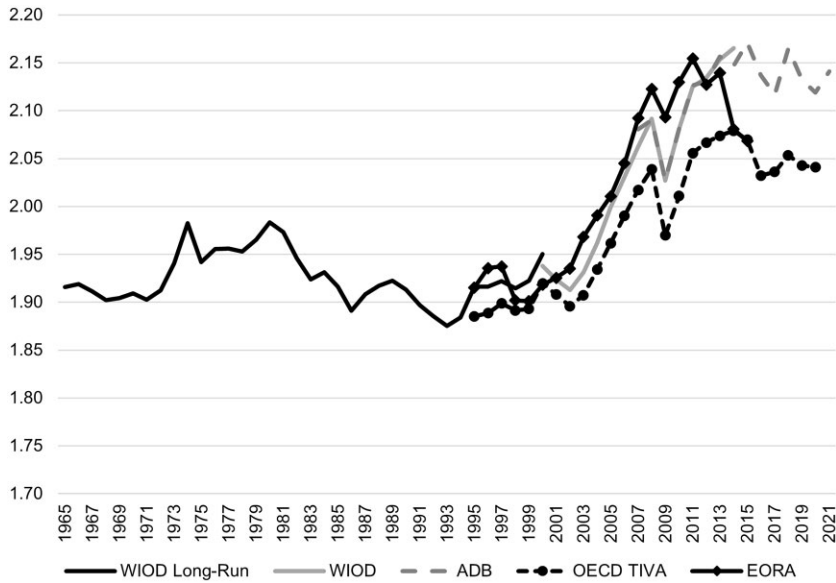
Before exploring the evolution of the various GVC positioning indicators, it is thus important to first investigate the issue of comparability of the different ICIO sources. Indeed, a substantial body of research has traditionally delved into this issue, employing various methodologies, such as direct comparison, sensitivity analysis, and decomposition analysis, with the goal of uncovering the underlying causes of disparities (Geschke et al. 2014; Inomata and Owen 2014; Moran and Wood 2014; Owen et al. 2014; Owen et al. 2016; Casella et al. 2019). These comprehensive investigations collectively demonstrate that disparities within ICIO datasets generally fall within a range of 10 percent for most values pertaining to the largest and structurally central economies and to around 30 percent for smaller economies or those with less comprehensive or reliable data (Casella et al. 2019).

For this paper's specific inquiry, attention is directed towards the matter of trade data comparability. In principle, ICIO data should conform to the trade data routinely used by economists and policymakers in the investigation of international trade in goods and services. This alignment is expected as both sources rely on common data origins. To ascertain this alignment and, in a roundabout manner, offer insights into the trustworthiness of the trade statistics provided in ICIO datasets, a comparative analysis is presented in table 1. This analysis specifically centers on overlapping countries and years, and directly compares the trade data in goods and services derived from our ICIO tables with equivalent data from the IMF World Economic Outlook (WEO), updated to October 2022.¹³ These comparisons are quantified as calculated discrepancies, considering relative percentage differences and annual growth rate differences of both export and import flows.

Some discrepancies with IMF trade data have been detected (see table 1), which might be attributed to the nature of ICIO's trade block representations at the sector-to-sector level, which are typically inferred or estimated (Casella et al. 2019). However, while disparities exist, they seem not to be substantial, both in terms of percentage difference in trade flows (top panel), in particular for import data, and growth

13 The analysis excluded the Long Run WIOD due to the limited temporal overlap with the other datasets. During this overlapping period, the data within the Long Run WIOD closely resemble those contained in the WIOD.

Figure 1. GVC Positioning over Time (World Average)



Source: Authors' elaboration using data from all datasets.

rates percent differences (bottom panel). Nevertheless, ICIO trade data consistently exhibit lower values compared to the IMF, as the reported differences are, on average, all positive in favor of IMF data. Specifically, across all ICIO data, there is evidence of fat tails in the distribution of discrepancies, with a notable tendency for higher underestimation of trade data at the uppermost echelons of the distribution. EORA trade data stands out as the only ICIO table where higher values are observed compared to IMF statistics.¹⁴ Finally, TiVA data exhibit the lowest variance across almost all the metrics.

Acknowledging these inherent discrepancies across the ICIO datasets, a first overview of the GVC positioning indicators at the global level from the various available ICIO sources is provided in [fig. 1](#). At the global aggregate level, upstreamness and downstreamness coincide and, as already pointed out, are a proxy for global production complexity ([Antràs and Chor 2019](#)). Overall, that GVC complexity has increased during the hyper-globalization phase (1995–2008), and that the same trend is common to all the datasets considered. This empirical evidence confirms that GVCs got longer as a result of a rise in cross-border intermediate sales and purchases ([Miller and Temurshoev 2017](#); [Wang et al. 2017](#); [Antràs and Chor 2019](#)). Before this phase, it seems that among the 25 countries covered by Long-Run WIOD—the vast majority being advanced economies—GVC complexity expanded in the 1970s while declining in the 1980s. Thanks to the use of ICIO datasets, it is evident from [fig. 1](#) that in the most recent years (post 2011) the globalization process has lost momentum, confirming that the global economy entered a “slowbalization” phase ([Antràs 2020](#)). Further analytical investigation is required for a comprehensive exploration of these trends. This study’s dataset holds the potential to significantly facilitate such research endeavors.

To provide further details, some primary descriptive statistics for the GVC positioning indicators computed using the proposed ICIO datasets are presented in [table 2](#). These statistics are again exclusively focused on the subset of countries that possess data spanning all the datasets within overlapping years.

14 To note that, in this comparative analysis, this study is specifically examining countries and years that are encompassed by all ICIO datasets. Consequently, any observed differences should not be ascribed to developing nations characterized by limited statistical data availability, which can be exclusively found in the EORA dataset.

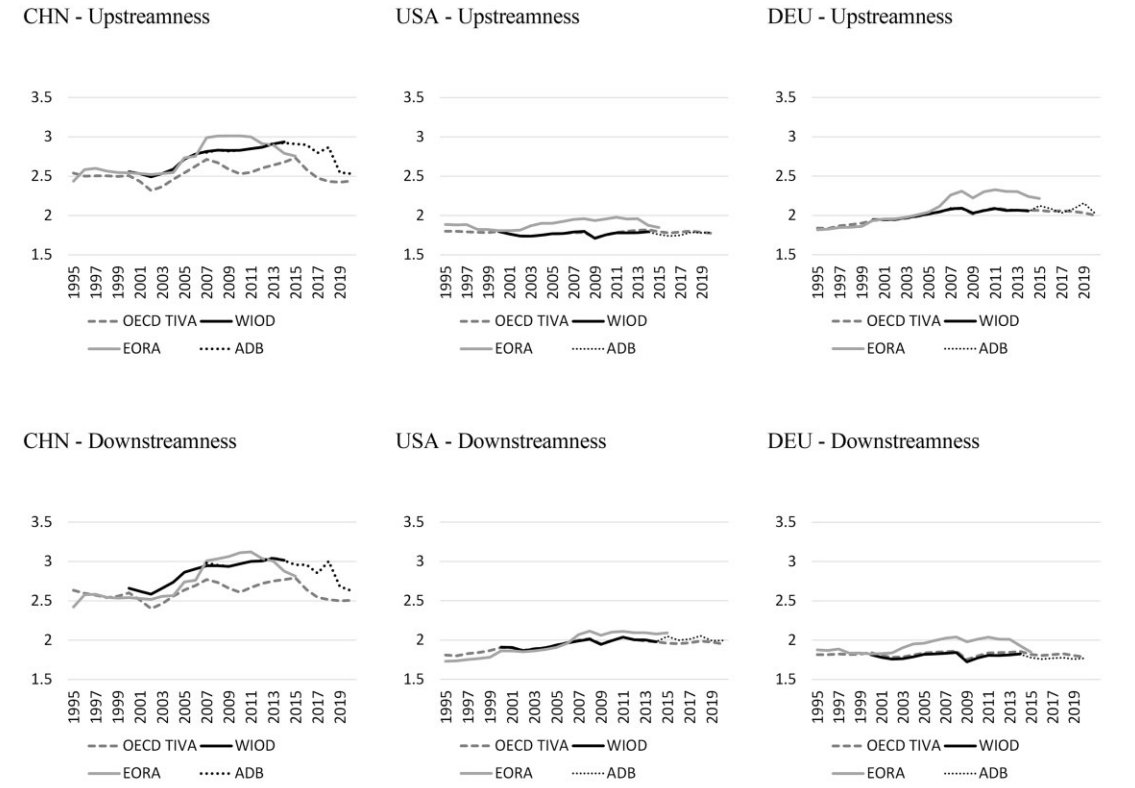
Table 2. Main Descriptive Statistics for the Global Value Chains' Positioning Indicators Computed Using the Reported ICIO Datasets (24 Overlapping Countries)

	Upstreamness					Downstreamness				
	Mean	st.d.	Min	Max	N	Mean	st.d.	Min	Max	N
1995–1999										
EORA	1.909	0.162	1.536	2.600	120	1.907	0.162	1.711	2.582	120
TIVA	1.896	0.217	1.634	2.539	120	1.906	0.238	1.708	2.635	120
LRWIOD	1.929	0.168	1.575	2.559	120	1.913	0.177	1.689	2.663	120
2000										
ADB	1.902	0.174	1.670	2.556	24	1.914	0.184	1.811	2.670	24
EORA	1.907	0.177	1.535	2.542	24	1.911	0.168	1.718	2.541	24
TIVA	1.920	0.233	1.652	2.505	24	1.943	0.260	1.727	2.601	24
LRWIOD	1.947	0.192	1.665	2.523	24	1.944	0.201	1.748	2.640	24
WIOD	1.902	0.174	1.672	2.555	24	1.912	0.182	1.813	2.659	24
2001–2006										
EORA	1.976	0.206	1.544	2.754	144	1.984	0.197	1.670	2.761	144
TIVA	1.938	0.253	1.627	2.627	144	1.968	0.274	1.721	2.694	144
WIOD	1.916	0.214	1.628	2.780	144	1.932	0.231	1.751	2.906	144
2007–2014										
ADB	2.033	0.353	1.672	2.921	192	2.069	0.386	1.726	3.047	192
EORA	2.135	0.349	1.534	3.284	192	2.146	0.357	1.688	3.180	192
TIVA	2.102	0.355	1.604	2.712	192	2.153	0.385	1.733	2.770	192
WIOD	2.033	0.353	1.675	2.934	192	2.066	0.385	1.724	3.039	192
2015										
ADB	2.096	0.437	1.757	2.909	24	2.113	0.445	1.778	2.957	24
EORA	2.088	0.353	1.575	3.114	24	2.098	0.350	1.671	2.956	24
TIVA	2.218	0.443	1.734	2.734	24	2.249	0.464	1.793	2.791	24
2016–2020										
ADB	2.067	0.373	1.701	2.898	120	2.087	0.405	1.759	3.003	120
TIVA	2.123	0.312	1.672	2.585	120	2.159	0.339	1.784	2.642	120
Total										
ADB	2.046	0.363	1.670	2.921	360	2.074	0.393	1.726	3.047	360
EORA	2.051	0.305	1.534	3.284	504	2.059	0.310	1.670	3.180	504
TIVA	2.065	0.332	1.604	2.734	624	2.103	0.359	1.708	2.791	624
LRWIOD	1.932	0.172	1.575	2.559	144	1.919	0.181	1.689	2.663	144
WIOD	1.991	0.314	1.628	2.934	360	2.017	0.343	1.724	3.039	360

Source: Authors' elaboration.

Across the different ICIO datasets, EORA registers, for the same subset of countries, a higher degree of structural complexity, representing the sole ICIO dataset with both indicators reaching three stages in terms of overall positioning after 2007. This result might be driven by the higher number of countries covered as counterparts for each reporting country, which makes the indicators more precise in measuring the actual number of cross-borders stages of production. The Long-run WIOD (LRWIOD) dataset exhibits the lowest degree of data dispersion around the mean (ranging between 0.17–0.18 overall). The actual values of the computed upstreamness and downstreamness indicators for each of the overlapping countries and time periods are reported in [table S1.2 in the supplementary online appendix](#). It should be noted that at the country level the degree of heterogeneity increases (e.g., as expected China is typically positioned about one step away from each endpoint in comparison to the United States across all

Figure 2. GVC Positioning Measures for China, Germany, and the United States: A Comparison across EORA, OECD TiVA, WIOD, and ADB



Source: Authors' elaboration.

the available datasets), although observed differences remain relatively modest as this study is looking at country-level aggregate measures.¹⁵ It is important to highlight that while these values do reflect nuanced distinctions, a discernible trend of relative consistency emerges in the positioning of various countries across the diverse datasets and time spans. To gain insight into this, a comparative analysis of the dynamics of GVC positioning indicators for three key countries within this common subset, namely China, Germany, and the United States, is presented in [fig. 2](#) across EORA, OECD TiVA, WIOD, and ADB datasets. Although the figures are different in detail, both the level gaps and the evolution of the time trends for each country look highly consistent across WIOD, OECD TiVA, and ADB, with the relevant exception of China.

Further support for this is provided by [table 3](#), which illustrates the significant level of correlation among upstreamness and downstreamness measures for the different ICIO datasets. Notably, all datasets exhibit a very strong correlation between them, as the lowest is the one between OECD TiVA, and EORA (slightly above 70 percent).

A more comprehensive examination of the temporal evolution of computed upstreamness and downstreamness indicators across the various world regions is presented in [tables 4](#) and [5](#). It is important to mention that, in this case, the data originate from different countries across the various ICIO sources,

15 It should also be noted that at the country level it was not possible to discern whether the observed rise in GVC positioning is a result of the overall expansion of GVCs or reflects the upgrading patterns of individual countries.

Table 3. Correlations of Indicators of Upstreamness and Downstreamness across the Different Inter-Country Input-Output Datasets

	Upstreamness					Downstreamness				
	ADB	EORA	TiVA	WIOD LR	WIOD	ADB	EORA	TiVA	WIOD LR	WIOD
ADB	1.000					1.000				
EORA	0.919	1.000				0.917	1.000			
TiVA	0.829	0.732	1.000			0.852	0.724	1.000		
WIOD LR	0.926	0.909	0.810	1.000		0.936	0.937	0.840	1.000	
WIOD	1.000	0.921	0.830	0.927	1.000	0.999	0.919	0.852	0.936	1.000

Source: Authors' elaboration.

making direct detailed comparisons even more challenging (see [table S1.3 in the supplementary online appendix](#) for the composition of geographical areas for each ICIO source).¹⁶ It is worth noting that the indicators have been weighted by the relative economic importance of each country within the same regional group, that is, by using the countries' positioning denominator values. As expected, by examining these indicators at the broader, regional level, the average GVC position is relatively consistent across regions, generally hovering around two stages, albeit with some variations. Notably, East Asian and Pacific countries exhibit the most substantial engagement in GVCs, both in terms of upstreamness and downstreamness. The same temporal evolution by income groups is reported in [tables 6 and 7](#) (refer to [table S1.4 in the supplementary online appendix](#) for the composition of income groups).¹⁷ To note that, on average, countries in the upper middle-income group are engaged in longer GVCs, both upstream and downstream, compared to the other groups throughout the entire period. High-income countries also exhibit a robust positive correlation with the percentage variation over time of both upstreamness and downstreamness for the entire period 1995–2015 (see [table S1.5 in the supplementary online appendix](#)).¹⁸ A snapshot of this evolution over time of both upstreamness and downstreamness, categorized by income group, is offered by [fig. 3](#). It uses the EORA data, which encompasses the largest subgroups of low-income countries. Looking at the position of countries with respect to the 45 degrees line, it is possible to identify which countries have changed their relative position in GVCs over the period. Both upstreamness and downstreamness plots show that, in general, high-income economies have shifted their production toward more “extreme” sectors: most of the countries below the 45° line—thus those countries that have experienced an increase in the positioning metrics—are high income. On the opposite, most of the countries above the 45° line are low-middle income economies, suggesting that such countries have shifted towards less upstream and less downstream sectors.

Additional insights for the empirical analysis can be gleaned from assessing the dynamics of the indicators by sectors. However, also in this respect, it is essential to acknowledge that the various ICIO datasets do not share a uniform sectoral classification and exhibit varying degrees of sectoral aggregation. Specifically, while WIOD, OECD TiVA, and ADB datasets can be aligned with the ISIC rev. 4 classification, EORA—standing out for its relatively highly aggregated sectors—can only be matched with ISIC rev.3. Consequently, this precludes direct alignment between EORA and any of the other datasets. Furthermore, to facilitate sectoral comparisons, it is necessary to harmonize all the datasets at the less granular level of disaggregation: for instance, to compare WIOD and OECD TiVA, it is

16 The geographical classification is taken from the World Bank and is available at the following link <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>.

17 The income classification is also taken from the World Bank and is available at the link reported in footnote 15.

18 The observation that countries and country-industries significantly distant from final demand also tend to be distant from the utilization of primary factors constitutes a “puzzling correlation,” already emphasized by [Antràs and Chor \(2019\)](#).

Table 4. Global Value Chains’ Upstreamness by Region over Time

Period	1995–1999				2000				2001–2006							
	EORA	TIVA	LRWIOD	ADB	EORA	TIVA	LRWIOD	WIOD	EORA	TIVA	WIOD					
Upstreamness—all countries																
East Asia and Pacific	2.004	2.066	2.020	2.051	2.040	2.101	2.045	2.034	2.150	2.175	2.121					
Europe and Central Asia	1.927	1.898	2.015	1.978	1.954	1.943	2.072	1.978	2.003	1.959	1.994					
Latin America and the Caribbean	1.806	1.760	1.804	1.746	1.807	1.790	1.809	1.758	1.852	1.813	1.785					
Middle East and Pacific	1.998	1.924		1.831	2.042	2.020		1.819	1.991	2.110	1.835					
North America	1.865	1.798	1.808	1.810	1.820	1.807	1.821	1.809	1.889	1.769	1.773					
South Asia	1.760	2.119	1.894	1.771	1.768	2.093	1.896	1.801	1.859	2.114	1.923					
Sub-Saharan Africa	1.956	2.081			1.974	2.265			1.990	2.211						
Total	1.918	1.913	1.930	1.922	1.918	1.940	1.949	1.916	1.986	1.961	1.933					
Period	2007–2014				2015				2016–2020				1995–2020			
Upstreamness—all countries																
East Asia and Pacific	2.400	2.457	2.418	2.402	2.519	2.420	2.570	2.420	2.372	2.408	2.309	2.353	2.024	2.311		
Europe and Central Asia	2.057	2.083	1.999	2.052	2.074	2.033	1.995	2.072	2.008	2.061	2.034	1.980	2.024	2.031		
Latin America and the Caribbean	1.803	1.907	1.850	1.830	1.803	1.843	1.796	1.812	1.823	1.804	1.874	1.823	1.805	1.816		
Middle East and Pacific	1.952	1.956	2.187	1.935	1.924	1.909	1.879	2.227	1.920	2.068	1.965	2.051		1.901		
North America	1.795	1.956	1.795	1.795	1.780	1.862	1.799	1.783	1.793	1.790	1.913	1.790	1.811	1.788		
South Asia	1.851	1.995	1.980	1.894	1.855	1.925	1.831	1.777	1.798	1.817	1.934	1.929	1.895	1.898		
Sub-Saharan Africa		1.978	2.170			1.899	2.016		2.032		1.971	2.123				
Total	2.058	2.118	2.109	2.053	2.109	2.068	2.201	2.081	2.121	2.066	2.048	2.073	1.934	2.010		

Source: Authors' elaboration.

Table 5. Global Value Chains' Downstreamness by Region Over Time

Period	1995–1999				2000				2001–2006							
	EORA	TIVA	LRWIOD	ADB	EORA	TIVA	LRWIOD	WIOD	EORA	TIVA	WIOD					
Downstreamness—all countries																
East Asia and Pacific	2.021	2.090	2.012	2.060	2.042	2.129	2.042	2.039	2.151	2.216	2.150					
Europe and Central Asia	1.911	1.884	1.978	1.950	1.946	1.929	2.044	1.947	1.984	1.942	1.956					
Latin America and the Caribbean	1.849	1.765	1.836	1.836	1.850	1.799	1.885	1.848	1.860	1.810	1.836					
Middle East and Pacific	1.917	1.818		2.243	1.892	1.843		2.207	1.839	1.834	2.151					
North America	1.869	1.820	1.791	1.824	1.834	1.848	1.818	1.824	1.926	1.819	1.801					
South Asia	1.751	2.185	1.981	1.863	1.738	2.164	1.980	1.883	1.843	2.224	2.009					
Sub-Saharan Africa	1.889	1.895			1.838	2.027			1.865	1.976						
Total	1.918	1.921	1.914	1.926	1.918	1.955	1.945	1.918	1.986	1.979	1.939					
Period	2007–2014				2015				2016–2020				1995–2020			
Downstreamness—all countries																
East Asia and Pacific	2.463	2.478	2.493	2.465	2.548	2.439	2.618	2.475	2.428	2.463	2.324	2.412	2.018	2.363		
Europe and Central Asia	2.007	2.044	1.984	2.005	2.018	1.996	1.965	2.011	1.983	2.007	2.004	1.962	1.989	1.987		
Latin America and the Caribbean	1.873	1.921	1.851	1.875	1.857	1.870	1.815	1.865	1.822	1.868	1.893	1.825	1.845	1.864		
Middle East and Pacific	2.434	1.826	1.853	2.419	2.414	1.791	1.751	2.427	1.750	2.425	1.836	1.808		2.337		
North America	1.818	2.007	1.836	1.818	1.799	1.906	1.823	1.780	1.812	1.802	1.951	1.824	1.797	1.812		
South Asia	2.010	1.993	2.150	2.042	1.983	1.930	1.926	1.956	1.896	1.981	1.929	2.053	1.981	2.031		
Sub-Saharan Africa		1.866	1.935			1.798	1.909		1.919		1.862	1.933				
Total	2.076	2.118	2.143	2.068	2.116	2.068	2.225	2.094	2.150	2.081	2.048	2.099	1.919	2.022		

Source: Authors' elaboration.

Table 6. Global Value Chains' Upstreamness by Income Group Over Time

Period	1995–1999						2000						2001–2006					
	EORA	TIVA	LRWIOD	ADB	EORA	TIVA	EORA	TIVA	LRWIOD	WIOD	EORA	TIVA	EORA	TIVA	WIOD	EORA	TIVA	WIOD
Upstreamness—all countries																		
High income	1.900	1.855	1.914	1.890	1.896	1.872	1.896	1.872	1.929	1.887	1.953	1.875	1.885					
Low income	1.934				1.927						1.968							
Lower-middle income	1.873	2.016	1.894	1.798	1.853	2.033	1.853	2.033	1.896	1.801	1.924	2.056	1.923					
Upper-middle income	2.033	2.123	2.120	2.190	2.065	2.163	2.065	2.163	2.169	2.171	2.176	2.209	2.265					
Total	1.916	1.913	1.930	1.922	1.916	1.940	1.916	1.940	1.949	1.916	1.984	1.961	1.933					
Period	2007–2014						2015						2016–2020					
Upstreamness—all countries	ADB	EORA	TIVA	WIOD	ADB	EORA	TIVA	EORA	TIVA	ADB	TIVA	ADB	EORA	TIVA	LRWIOD	WIOD		
High income	1.935	2.044	1.924	1.929	1.929	1.985	1.912	1.985	1.912	1.927	1.912	1.930	1.989	1.901	1.917	1.912		
Low income		2.004				1.992							1.987					
Lower-middle income	1.891	1.985	1.983	1.894	1.899	1.917	1.863	1.899	1.863	1.829	1.857	1.862	1.953	1.943	1.895	1.898		
Upper-middle income	2.483	2.382	2.394	2.488	2.604	2.315	2.540	2.604	2.540	2.498	2.359	2.495	2.299	2.355	2.130	2.435		
Total	2.058	2.117	2.109	2.053	2.109	2.067	2.201	2.109	2.201	2.081	2.121	2.066	2.047	2.073	1.934	2.010		

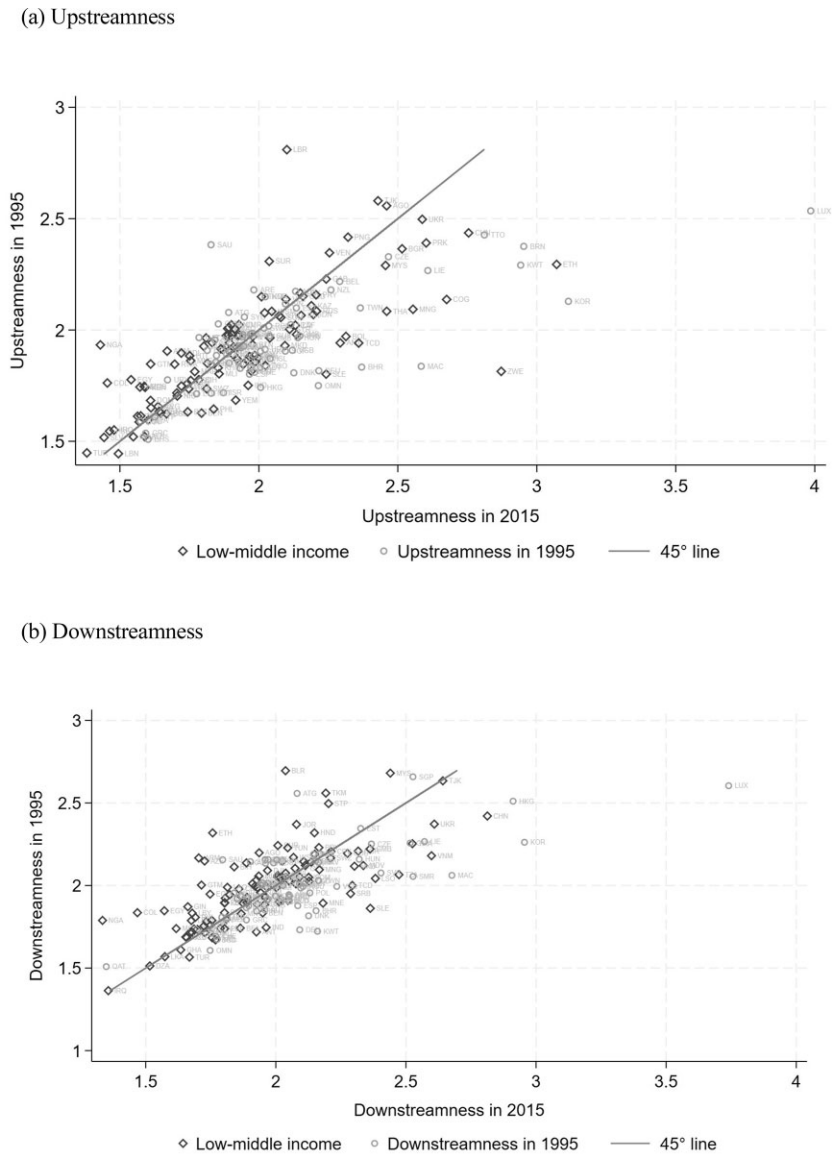
Source: Authors' elaboration.

Table 7. Global Value Chains’ Downstreamness by Income Group Over Time

Period	1995–1999						2000						2001–2006											
	EORA	TIVA	LRWIOD	ADB	EORA	TIVA	LRWIOD	WIOD	EORA	TIVA	WIOD	EORA	TIVA	WIOD										
Downstreamness—all countries																								
High income	1.903	1.857	1.889	1.888	1.901	1.879	1.911	1.883	1.963	1.885	1.887													
Low income	1.909				1.839				1.916															
Lower-middle income	1.853	2.069	1.981	1.887	1.791	2.065	1.980	1.883	1.867	2.105	2.009													
Upper-middle income	2.035	2.147	2.180	2.221	2.062	2.202	2.268	2.205	2.150	2.244	2.292													
Total	1.918	1.921	1.914	1.926	1.918	1.955	1.945	1.918	1.986	1.979	1.939													
Period	2007–2014						2015						2016–2020						1995–2020					
Downstreamness—all countries	ADB	EORA	TIVA	TIVA	WIOD	ADB	EORA	TIVA	ADB	TIVA	ADB	EORA	TIVA	LRWIOD	WIOD									
High income	1.938	2.047	1.935	1.932	1.924	1.988	1.909	1.905	1.904	1.923	1.994	1.972	1.905	1.892	1.915									
Low income		2.001				1.994																		
Lower-middle income	2.065	1.945	2.080	2.042	2.043	1.883	1.949	2.033	1.966	2.045	1.912	2.033	1.981	2.031										
Upper-middle income	2.520	2.395	2.452	2.525	2.616	2.323	2.584	2.551	2.412	2.537	2.303	2.406	2.197	2.470										
Total	2.076	2.119	2.143	2.068	2.116	2.069	2.225	2.094	2.150	2.081	2.049	2.099	1.919	2.022										

Source: Authors’ elaboration.

Figure 3. GVC Positioning Measures by Income Group and Their Correlation over Time–1995 and 2015

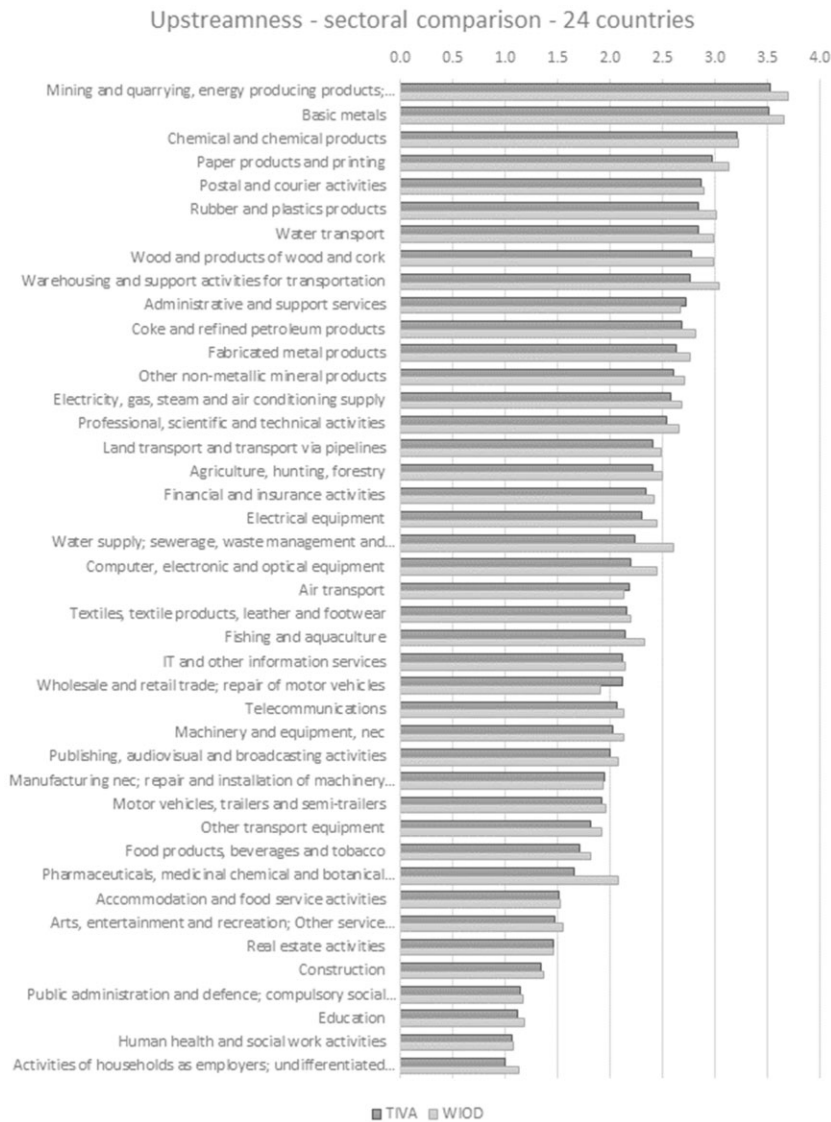


Source: Authors' elaboration using EORA26 data.

necessary to aggregate the 55 sectors from WIOD into the 45 sectors of OECD TiVA (in some circumstances further aggregated), and the same harmonization process applies to other dataset comparisons. A comprehensive breakdown of the comparison involving GVC positioning indicators for overlapping countries and periods, segmented by sectors according to the ISIC rev 4 classification, is provided in [tables S1.6–S1.11 in the supplementary online appendix](#). A snapshot of sectoral comparisons, specifically focusing on WIOD and OECD TiVA datasets, is provided in [fig. 4](#).

To begin, a noteworthy level of consistency in the GVC positioning indicators between the two datasets is also observed at the sectoral level. This indicates that the sectors considered relatively upstream and/or downstream align closely, albeit with WIOD generally exhibiting higher indicators compared to OECD

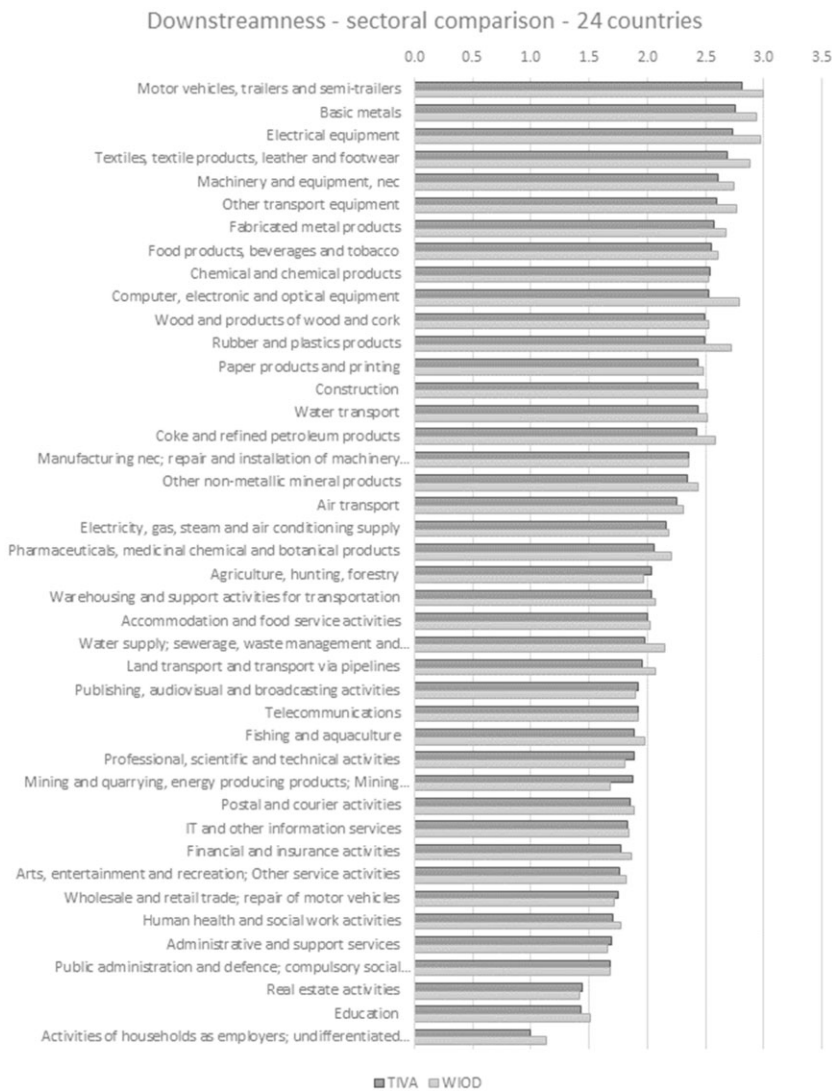
Figure 4. GVC Upstreamness and Downstreamness by Sector (World Average, 2000–2014)



Source: Authors' elaboration.

TiVA overall. Furthermore, as expected, manufacturing sectors are, on average, characterized by longer GVCs than services. All manufacturing sectors exhibit greater chain length in relation to both endpoints of the production chains, typically indicating an average of two stages of production away from each endpoint. This trend remains consistent across manufacturing sectors, exhibiting relatively limited cross-industry variation. In both datasets, the most upstream sectors are Mining and Quarrying, and Basic Metals, two sectors usually involved in the first stages of global value chains. Basic Metals are also listed alongside Motor Vehicles and Electrical Equipment as some of the most downstream industries. This is again consistent with [Antràs and Chor's \(2019\)](#) puzzling correlation and, in this case, is also influenced by the relatively high level of sector aggregation.

Figure 4 – continued



5. Conclusions

The availability of new indicators of GVCs positioning at the country and sectoral levels provides an unprecedented opportunity to carry out qualitative and quantitative analyses on different economic aspects related to GVCs. For the sake of future use by other scholars, this work computes and provides access to a new and constantly updated dataset of GVC positioning indicators at the country, country-industry, industry, and aggregate levels, based on the most used global ICIO tables.

Despite the inherent disparities in the characteristics and computation methods of the various ICIO datasets, which prevent straightforward comparisons of the computed GVC position indicators, this paper offers informative insights on these metrics. These encompass overlapping countries and periods, sectors, geographical regions, and income levels. Overall, while acknowledging the existing differences, a commendable degree of consistency among the GVC positioning measures across the diverse ICIO datasets has been observed. Considering this, refraining from attributing inherent superiority to any one

database over the others is essential, as the choice among datasets depends on specific research objectives. Researchers seeking the broadest country coverage, particularly in the context of developing economies, may find the EORA database most suitable. Those placing emphasis on data holding official status and a more refined sectoral breakdown could opt for WIOD, which is, however, not as up to date as the OECD TiVA and ADB datasets. For an historical perspective, the Long-run WIOD seems the most suited.

The dataset presented in this paper possesses the potential to significantly facilitate research endeavors related to the analyses of GVCs and production networks. The authors are confident that scholars will find value in this work and use the provided indicators to enhance their investigations into GVCs. Furthermore, these insights are expected to prove beneficial for policymakers in the pursuit of informed decision-making processes.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this article.

Data Availability Statement

The global dataset of GVC upstreamness and downstreamness measures presented in this article is available at the following link <https://www.tradeconomics.com/position/>. Codes to replicate the dataset, as well as to compute positioning measures on other ICIO tables, are available here: https://www.tradeconomics.com/position/docs/position_codes.zip.

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Supplementary Online Appendix
**Positioning in Global Value Chains: World Map and Indicators, a New
Dataset Available for GVC Analyses**
M. Mancini, P. Montalbano, S. Nenci , and D. Vurchio

Additonal Tables
Table S1.1

Table S1.1. Main Characteristics of the ICIO Datasets Used to Compute GVC Positioning Measures

Dataset	Countries	Industries	Years	Data sources	References	Website
EORA	189 countries	26 industries (in the EORA26 version, ISIC - Rev. 3)	1990-2015 (updates for more recent years are available but subject to licensing)	Input-output tables, main aggregate data from national statistical offices, UN National Accounts	Lenzen et al, 2012 ; and Lenzen et al, 2013	https://www.worldmrio.com
TiVA	76 countries (2022 edition)	45 industries (ISIC Rev. 4)	1995-2020	OECD Inter-Country Input-Output (ICIO)	OECD-WTO (2012) “Trade in Value Added: Concepts, Methodologies and Challenges”, Joint OECD-WTO concept note.	https://www.oecd.org/sti/ind/measuring-trade-in-value-added.htm
WIOD	43 countries (2016 release)	56 industries (ISIC Rev. 4)	2000-2014	National Supply-Use tables	Timmer et al., 2015	https://www.rug.nl/ggdc/valuechain/wiod/
Long-run WIOD	25 countries	23 sectors (ISIC Rev. 3.1)	1965-2000	National Input-Output tables	Woltjer et al., 2021	https://www.rug.nl/ggdc/valuechain/long-run-wiod
ADB MRIO	62 countries (with a focus on Asian economies)	35 sectors (ISIC Rev. 3.1)	2000, 2007-2022	An extension of WIOD (the information for the additional Asian countries is derived from estimations produced by researchers, not from official statistics)		https://kidb.adb.org/mrio

Source: Authors' elaboration.

Table S1.3a. Continued

ADB	East Asia and Pacific				Europe and Central Asia				South Asia			
	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA
	Taiwan				Poland	Iceland	Norway		Romania			
	Thailand				Portugal	Ireland	Poland		Russian Fed.			
	Vanuatu				Romania	Italy	Portugal		Slovakia			
	Viet Nam				Russian F.		Romania		Slovenia			
					Slovakia	Kazakhstan	Russian Federation		Spain			
					Slovenia	Kyrgyzstan	Slovakia		Sweden			
					Spain	Latvia	Slovenia					
					Sweden	Liechtenstein			Switzerland			
					Switzerland	Lithuania	Spain		Turkey			
					Turkey	Luxembourg	Sweden		United Kingdom			
					UK	Monaco						
							Switzerland					
						Montenegro	Turkey					
							Ukraine					
						Netherlands						
						Norway	United Kingdom					
						Poland						
						Portugal						
						Rep. of						
						Moldova						
						Romania						
						Russian Federation						
						San Marino						
						Serbia						
						Slovakia						
						Slovenia						
						Spain						
						Sweden						
						Switzerland						
						Tajikistan						
						TPYR of Macedonia						
						Turkey						
						Turkmenistan						
						Ukraine						
						United Kingdom						
						Uzbekistan						

Table S1.3b. Composition of geographical regions by ICIO source.

ADB	Middle East and Pacific				North America				Latin America and the Caribbean					
	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA	LRWIOD	WIOD
Malta	Algeria	Egypt		Malta	Canada	Canada	Canada	Canada	Canada	Brazil	Antigua and Barbuda	Argentina	Brazil	Brazil
	Bahrain	Israel												
	Djibouti	Jordan			USA		USA	USA	USA	Mexico	Argentina	Brazil	Mexico	
	Egypt	Malta			ADB	EORA	Sub-Saharan Africa	LRWIOD	WIOD		Aruba	Chile		Mexico
	Iran	Morocco				Angola	TIVA				Bahamas	Colombia		
											Barbados	Costa Rica		
	Iraq	Saudi Arabia				Benin	Cameroon				Belize	Mexico		
	Israel	Tunisia				Botswana	Côte d'Ivoire				Bolivia	Peru		
	Jordan					Burkina Faso	Nigeria				Bt Virgin Islds			
	Kuwait					Burundi	Senegal				Brazil			
	Lebanon						South Africa							
	Libya					Cabo Verde					Chile			
	Malta													
	Morocco					Cameroon								
	Oman					Central African Rep.					Costa Rica			
	Qatar					Chad					Cuba			
	Saudi Arabia					Congo								
						Côte d'Ivoire					Dominican Rep.			
						Dem. Rep. of the Congo					Ecuador			
						Eritrea					El Salvador			
						Eswatini					Guatemala			
						Ethiopia					Guyana			
						Gabon					Haiti			
											Honduras			
						Gambia					Jamaica			
						Ghana					Mexico			
						Guinea								
											Netherlands Ant.			
						Kenya								
											Nicaragua			

Table S1.3b. Continued

ADB	Middle East and Pacific				North America				Latin America and the Caribbean			
	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA
						Lesotho Liberia					Panama Paraguay Peru	
						Madagascar Malawi Mali					Suriname Trinidad and Tobago Uruguay	
						Mauritania Mauritius					Venezuela	
						Mozambique Namibia Niger Nigeria Rwanda Sao Tome and Principe Senegal Seychelles						
						Sierra Leone Somalia South Africa South Sudan Sudan Togo Uganda United Rep. of Tanzania Zambia						
						Zimbabwe						

Table S1.4a. Composition of income groups by IClO source (WB classification).

ADB	EORA	Low income			Lower middle income			WIOD	
		TIVA	LRWIOD	WIOD	ADB	EORA	TIVA		LRWIOD
	Alghanistan				Bangladesh	Algeria	Bangladesh		
	Burkina Faso				Bhutan	Angola	Cambodia		India
	Burundi				Cambodia	Bangladesh	Cameroon		
	Central African				India	Benin	Côte d'Ivoire		
	Rep.								
	Chad				Kyrgyzstan	Bhutan	Egypt		
	Dem. Rep. of the				Laos	Bolivia	India		
	Congo								
	Eritrea				Mongolia	Cabo Verde	Jordan		
	Ethiopia				Nepal	Cambodia	Laos		
	Gambia				Pakistan	Cameroon	Morocco		
	Liberia				Philippines	Congo	Myanmar		
	Madagascar				Sri Lanka	Côte d'Ivoire	Nigeria		
	Malawi				Viet Nam	Djibouti	Pakistan		
	Mali					Egypt	Philippines		
	Mozambique					Eswatini	Senegal		
	N.Korea					Ghana	Tunisia		
	Niger					Guinea	Ukraine		
	Rwanda					Haiti	Viet Nam		
	Sierra Leone					Honduras			
	Somalia				India	India			
	South Sudan				Iran				
	Sudan				Jordan				
	Syria				Kenya				
	Togo				Kyrgyzstan				
	Uganda				Lao People's				
	Yemen				Dem. Rep.				
					Lebanon				
					Lesotho				
					Mauritania				
					Mongolia				
					Morocco				
					Myanmar				
					Nepal				
					Nicaragua				
					Nigeria				
					Pakistan				
					Papua New				
					Guinea				
					Philippines				
					Samoa				
					Sao Tome and				
					Principe				
					Senegal				
					Sri Lanka				
					Tajikistan				
					Tunisia				
					Ukraine				
					United Rep. of				
					Tanzania				
					Uzbekistan				
					Vanuatu				
					Viet Nam				
					Zambia				
					Zimbabwe				

Table S1.4b. Composition of income groups by ICI source (WB classification).

ADB	Upper middle income				High income			
	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA	LRWIOD
								WIOD
Brazil	South Africa	South Africa	China	Mexico	Australia	Andorra	Australia	USA
Fiji	TFYR of Macedonia	Bulgaria	Brazil	China	Austria	Antigua & Barbuda	Austria	Australia
Indonesia	Turkmenistan	Kazakhstan	Mexico	Indonesia	Belgium	Aruba	Belgium	Austria
Turkey	Albania	Indonesia		Bulgaria	Brunei	Australia	Brunei	Belgium
					Darussalam		Darussalam	Canada
Thailand	Belarus	Colombia		Brazil	Canada	Austria	Canada	Canada
Mexico	Peru	Thailand		Turkey	Hong Kong SAR	Bahamas	Chile	Denmark
Russian Federation	Gabon	Belarus		Russian Fed.	Croatia	Bahrain	Croatia	Finland
China	Mauritius	China			Cyprus	Barbados	Cyprus	France
Kazakhstan	Montenegro	Costa Rica			Czechia	Belgium	Czechia	Germany
Bulgaria	Iraq	Turkey			Denmark	Bermuda	Denmark	Greece
Malaysia	Fiji	Peru			Estonia	Br. Virgin Isds	Estonia	Hong Kong SAR
Maldives	Serbia	Russian Federation			Finland	Brunei	Finland	Ireland
					France	Darussalam		Italy
	State of Palestine	Malaysia			Germany	Canada	France	Italy
	Kazakhstan	Mexico			Greece	Cayman Isds	Germany	Japan
	Ecuador	Brazil			Hungary	Chile	Greece	Korea Rep. of
	Malaysia	Argentina			Ireland	Croatia	Hong Kong SAR	Netherlands
	Brazil				Italy	Cyprus	Hungary	Portugal
	Rep. of Moldova				Japan	Czechia	Iceland	Spain
	Russian Federation					Denmark	Ireland	Sweden
	Belize				Latvia	Estonia	Israel	Taiwan
	Turkey				Lithuania	Finland	Italy	United Kingdom
	China				Luxembourg	France	Japan	
	Georgia				Malta	French Polynesia	Korea Rep. of	
	Guatemala				Netherlands	Germany	Latvia	Netherlands
	Argentina				Norway	Greece	Lithuania	Norway
	Libya				Poland	Greenland	Luxembourg	Poland
	Maldives				Portugal	Guyana	Malta	Portugal
	El Salvador				Korea Rep. of	Hong Kong SAR	Netherlands	Romania
	Indonesia				Romania	Hungary	New Zealand	Slovakia
	Botswana				Singapore	Iceland	Norway	Slovenia
	Armenia				Slovakia	Ireland	Poland	Spain
	Mexico				Slovenia	Israel	Portugal	Sweden
	Bosnia				Spain	Italy	Romania	Switzerland
	Herzegovina				Sweden	Japan	Saudi Arabia	Taiwan
	Thailand				Switzerland	Kuwait	Singapore	United Kingdom
								USA

Table S1.4b. Continued

Upper middle income			High income						
ADB	EORA	TIVA	LRWIOD	WIOD	ADB	EORA	TIVA	LRWIOD	WIOD
	Suriname Costa Rica Colombia Paraguay Jamaica Bulgaria Azerbaijan Cuba Dominican Rep.		Taiwan United Kingdom USA			Larvia Liechtenstein Lithuania Luxembourg Macao SAR Malta Monaco Netherlands New Caledonia New Zealand Norway Oman Panama Poland Portugal Qatar Korea Rep. Of Romania San Marino Saudi Arabia Seychelles Singapore Slovakia Slovenia Spain Sweden Switzerland Taiwan Trinidad and Tobago United Arab Emirates United Kingdom Uruguay USA	Slovakia Slovenia Spain Sweden Switzerland Taiwan United Kingdom USA		

Table S1.11. Downstreamness by ISIC rev. 4 Sectors: A Comparison between ADB and OECD TIVA for Overlapping Countries and Periods

sectors	2000		2007-2014		2015		2016-2020		2000-2020	
	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA
1	1.907	1.899	1.991	2.066	1.997	2.090	1.904	2.071	1.953	2.021
2	1.796	1.836	1.655	1.833	1.698	2.013	1.857	1.956	1.733	1.879
3	2.434	2.419	2.664	2.602	2.716	2.625	2.652	2.593	2.655	2.550
4	2.502		2.942		3.024		2.953		2.934	
5	2.716	2.499	3.194	2.756	3.335	2.913	3.151	2.789	3.172	2.685
6	2.408	2.466	2.614	2.555	2.672	2.573	2.657	2.480	2.624	2.497
7,26,30	1.843	1.866	1.903	1.921	1.930	1.950	1.894	1.893	1.899	1.886
	Pulp, paper, paper products, printing, and publishing; Other supporting and auxiliary transport activities; activities of travel agencies; Renting of M&Eq and other business activities									
6	2.323	2.141	2.518	2.338	2.505	2.368	2.470	2.286	2.493	2.263
10	2.490	2.367	2.768	2.588	2.772	2.551	2.771	2.523	2.758	2.496
11	2.338	2.282	2.492	2.409	2.588	2.469	2.489	2.373	2.490	2.348
12	2.458	2.360	2.897	2.665	3.002	2.651	2.881	2.714	2.883	2.596
13,16	2.352	2.294	2.698	2.618	2.801	2.690	2.706	2.588	2.696	2.523
14	2.497	2.393	2.995	2.681	3.091	2.733	2.984	2.684	2.976	2.587
15	2.663	2.603	3.024	2.841	3.117	2.851	3.020	2.803	3.018	2.761
17,34	1.877	1.823	1.993	1.951	2.016	1.949	1.996	1.904	1.992	1.896
	Electricity, gas, and water supply; Other community, social, and personal services									
18	2.218	2.210	2.625	2.503	2.747	2.659	2.685	2.525	2.646	2.440
	Construction									

Table S1.11. Continued

sectors	2000		2007-2014		2015		2016-2020		2000-2020	
	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA
19	1.741		1.823		1.816		1.758		1.796	
20	1.712		1.762		1.742		1.742		1.751	
21	1.605	1.685	1.692	1.774	1.690	1.771	1.692	1.792	1.689	1.752
22	1.966	1.920	2.062	2.032	2.076	2.012	2.059	2.025	2.058	1.999
23	1.913	1.820	2.117	2.007	2.123	2.001	2.104	2.003	2.106	1.951
24	2.376	2.301	2.573	2.466	2.610	2.460	2.616	2.486	2.582	2.431
25	2.185	2.214	2.400	2.356	2.351	2.205	2.307	2.204	2.354	2.254
27	1.883	1.924	1.930	1.958	1.969	1.980	1.985	1.931	1.951	1.917
28	1.846	1.785	1.832	1.824	1.779	1.770	1.808	1.744	1.820	1.772
29	1.415	1.413	1.406	1.452	1.401	1.469	1.446	1.450	1.422	1.446
31	1.640	1.641	1.704	1.713	1.700	1.701	1.692	1.690	1.697	1.686
32	1.441	1.403	1.547	1.448	1.620	1.435	1.691	1.443	1.608	1.430
33	1.730	1.660	1.824	1.722	1.883	1.755	1.921	1.729	1.865	1.707
35	1.159	1.000	1.126	1.000	1.126	1.000	1.065	1.000	1.106	1.000
Total	1.914	1.895	2.069	2.021	2.113	2.058	2.087	2.014	2.074	1.981

Source: Authors' computations.

Note: Sectors' positioning measures are weighted according to their relative gross output.

Table S1.2. GVC Positioning Indicators (Upstreamness and Downstreamness) across Time Periods (Only for Overlapping Countries and Periods)

Period	1995-1999				2000				2001-2006				2007-2014				2015				2016-2020			
	EORA	TIVA	WIOD LR	ADB	EORA	TIVA	WIOD LR	WIOD	EORA	TIVA	WIOD	ADB	EORA	TIVA	WIOD	ADB	EORA	TIVA	ADB	TIVA	ADB	TIVA		
Upstreamness	AUS	2.181	2.009	2.189	2.196	2.125	2.058	2.239	2.193	2.246	2.101	2.187	2.247	2.190	2.165	2.256	2.197	2.132	2.112	2.296	2.209			
	AUT	1.861	1.801	1.948	1.900	1.905	1.863	2.012	1.901	1.972	1.919	1.945	2.049	2.062	2.009	2.044	2.030	2.006	1.998	2.036	1.997			
	BEL	2.222	2.089	2.178	2.131	2.232	2.146	2.285	2.134	2.296	2.147	2.126	2.179	2.357	2.141	2.174	2.184	2.291	2.126	2.178	2.134			
	BRA	1.846	1.713	1.848	1.822	1.931	1.839	1.937	1.846	2.008	1.941	1.920	1.866	2.060	1.890	1.899	1.818	1.998	1.827	1.862	1.865			
	CAN	1.985	1.882	1.998	2.005	1.998	1.971	2.088	2.009	2.065	1.941	1.986	1.996	2.047	1.912	2.000	2.068	1.982	1.859	1.980	1.867			
	CHN	2.546	2.509	2.510	2.556	2.542	2.505	2.523	2.555	2.603	2.457	2.607	2.854	2.953	2.621	2.857	2.909	2.755	2.734	2.727	2.471			
	DEU	1.840	1.865	1.920	1.947	1.938	1.936	1.988	1.947	2.009	1.983	1.988	2.070	2.283	2.069	2.068	2.123	2.216	2.065	2.076	2.039			
	DNK	1.824	1.821	1.912	1.848	1.904	1.888	2.012	1.847	1.928	1.888	1.875	1.934	2.143	1.897	1.929	1.927	2.126	1.874	1.954	1.881			
	ESP	1.827	1.816	2.045	1.841	1.858	1.852	2.051	1.840	1.946	1.907	1.935	1.932	1.968	1.876	1.928	1.968	1.966	1.861	1.943	1.864			
	FIN	2.126	2.026	2.214	2.132	2.139	2.097	2.305	2.140	2.167	2.038	2.091	2.073	2.171	1.994	2.073	2.100	2.096	1.960	2.092	1.983			
	FRA	1.951	1.824	1.964	1.896	1.960	1.872	2.030	1.895	1.966	1.864	1.885	1.884	1.974	1.846	1.880	1.904	1.885	1.847	1.919	1.862			
	GBR	1.973	1.835	2.098	1.932	1.949	1.851	2.116	1.933	1.963	1.850	1.920	1.948	2.005	1.887	1.943	1.919	1.970	1.853	1.910	1.861			
	GRC	1.541	1.667	1.605	1.717	1.535	1.652	1.665	1.718	1.555	1.651	1.726	1.767	1.573	1.685	1.765	1.843	1.593	1.734	1.849	1.704			
	IND	1.747	2.248	1.896	1.799	1.757	2.205	1.896	1.801	1.849	2.219	1.913	1.895	2.036	2.065	1.900	1.889	1.961	1.876	1.810	1.834			
	IRL	1.852	2.283	2.172	2.075	1.894	2.288	2.238	2.076	1.905	2.205	2.073	2.110	1.934	2.266	2.106	2.084	1.893	2.282	2.154	2.230			
	ITA	1.969	1.945	2.081	1.984	1.949	1.978	2.142	1.982	1.968	1.973	1.991	2.002	1.989	1.941	1.997	1.981	1.942	1.926	1.968	1.956			
	JPN	1.833	1.766	1.881	1.877	1.867	1.762	1.877	1.875	1.885	1.776	1.884	1.894	1.961	1.827	1.887	1.853	1.926	1.840	1.861	1.833			
	KOR	2.181	2.030	2.222	2.096	2.154	2.047	2.270	2.092	2.314	2.029	2.107	2.294	3.130	2.189	2.292	2.395	3.114	2.207	2.314	2.179			
	MEX	1.613	1.785	1.740	1.670	1.587	1.764	1.694	1.672	1.614	1.732	1.652	1.702	1.642	1.771	1.707	1.779	1.575	1.739	1.731	1.755			
	NLD	2.043	2.028	2.148	2.126	2.064	2.067	2.186	2.129	2.131	2.060	2.045	2.151	2.180	2.079	2.151	2.327	2.115	2.091	2.361	2.085			
PRT	1.834	1.895	2.031	1.839	1.877	1.874	2.018	1.839	1.850	1.850	1.864	1.919	1.896	1.834	1.918	1.927	1.933	1.809	1.956	1.819				
SWE	1.980	1.983	2.146	2.007	1.986	2.013	2.179	2.013	2.039	1.994	2.011	2.070	2.098	2.027	2.072	2.082	2.018	1.991	2.118	1.988				
TWN	2.074	2.015	2.087	1.991	2.177	2.017	2.106	1.992	2.242	2.052	2.064	2.262	2.386	2.232	2.268	2.360	2.365	2.239	2.271	2.186				
USA	1.859	1.793	1.796	1.796	1.807	1.796	1.803	1.796	1.870	1.754	1.755	1.773	1.946	1.782	1.773	1.757	1.847	1.794	1.768	1.787				

Table S1.2. Continued

Period	1995-1999				2000				2001-2006				2007-2014				2015				2016-2020			
	EORA	TIVA	WIOD	LR	ADB	EORA	TIVA	WIOD	LR	WIOD	EORA	TIVA	WIOD	ADB	EORA	ADB	TIVA	EORA	ADB	TIVA	ADB	TIVA	ADB	TIVA
Downstreamness																								
AUS	2.152	1.946	2.119		2.069	2.066	1.997	2.128		2.068	2.19	2.04	2.09	2.064	2.055	2.065	2.056	1.991	1.967	2.048	1.986			
AUT	1.876	1.796	1.851		1.858	1.895	1.836	1.900		1.857	1.94	1.89	1.91	2.001	2.028	1.998	1.958	1.958	1.970	2.036	1.971			
BEL	2.159	2.055	2.140		2.168	2.100	2.223	2.075		2.075	2.19	2.08	2.04	2.124	2.224	2.117	2.225	2.152	2.082	2.211	2.095			
BRA	1.888	1.730	1.853		1.851	1.966	1.867	1.966		1.873	2.02	1.92	1.90	1.895	2.076	1.892	1.893	1.850	2.018	1.838	1.852	1.826		
CAN	1.999	1.836	1.901		1.929	1.953	1.896	1.974		1.929	2.00	1.88	1.92	1.952	2.019	1.873	1.955	2.066	1.999	1.870	1.943	1.866		
CHN	2.532	2.580	2.597		2.670	2.541	2.601	2.640		2.659	2.61	2.54	2.73	2.989	3.032	2.710	2.983	2.957	2.813	2.791	2.825	2.542		
DEU	1.754	1.829	1.888		1.914	1.862	1.906	1.968		1.911	1.89	1.91	1.91	1.999	2.090	1.997	1.996	2.048	2.092	1.960	2.011	1.967		
DNK	1.828	1.814	1.899		1.867	1.916	1.840	1.948		1.865	1.95	1.87	1.90	1.971	2.154	1.929	1.969	1.970	2.126	1.902	1.959	1.903		
ESP	1.930	1.903	2.056		1.961	2.003	1.980	2.076		1.962	2.07	2.04	2.04	2.023	2.113	1.956	2.019	2.051	2.084	1.893	1.982	1.898		
FIN	2.005	1.900	2.020		1.997	1.996	1.960	2.083		1.993	2.01	1.94	1.98	2.050	1.999	1.992	2.040	2.035	1.946	1.946	2.045	1.972		
FRA	1.942	1.815	1.916		1.907	1.968	1.876	1.998		1.900	1.94	1.87	1.90	1.931	1.935	1.902	1.926	1.941	1.851	1.886	1.945	1.899		
GBR	1.995	1.817	2.077		1.898	1.970	1.827	2.105		1.896	1.99	1.83	1.89	1.932	2.050	1.873	1.929	1.876	1.970	1.847	1.880	1.845		
GRC	1.801	1.793	1.713		1.811	1.825	1.805	1.748		1.813	1.83	1.78	1.79	1.770	1.889	1.774	1.771	1.845	1.887	1.793	1.830	1.808		
IND	1.737	2.309	1.983		1.883	1.727	2.268	1.980		1.883	1.83	2.32	2.00	2.045	2.030	2.234	2.047	2.006	1.963	1.953	1.973	1.914		
IRL	2.183	2.325	2.214		2.074	2.154	2.274	2.221		2.067	2.15	2.26	2.07	2.089	2.239	2.314	2.079	1.978	2.170	2.231	1.995	2.240		
ITA	2.004	1.966	2.045		2.014	2.002	2.034	2.124		2.012	2.01	2.03	2.01	2.058	2.009	2.042	2.052	2.016	1.948	1.996	1.997	1.997		
JPN	1.818	1.740	1.861		1.861	1.844	1.727	1.855		1.857	1.85	1.75	1.88	1.929	1.916	1.840	1.923	1.899	1.926	1.831	1.885	1.819		
KOR	2.286	2.077	2.246		2.161	2.255	2.089	2.316		2.155	2.37	2.11	2.20	2.428	3.027	2.329	2.426	2.462	2.956	2.249	2.324	2.221		
MEX	1.727	1.799	1.811		1.821	1.718	1.784	1.815		1.824	1.70	1.76	1.78	1.837	1.728	1.817	1.848	1.868	1.671	1.793	1.883	1.814		
NLD	1.923	1.935	2.036		1.955	1.952	1.967	2.015		1.946	2.05	1.94	1.92	2.010	2.105	1.994	2.005	2.089	2.051	2.027	2.095	2.028		
PRT	2.019	2.066	2.140		2.057	2.005	2.054	2.143		2.061	2.05	2.00	2.02	1.992	2.071	1.975	1.994	1.941	2.028	1.915	1.982	1.927		
SWE	1.949	1.614	2.019		1.940	1.994	1.936	2.060		1.938	2.00	1.93	1.92	1.949	2.049	1.967	1.947	1.943	1.966	1.927	1.969	1.934		
TWN	1.989	2.005	2.058		2.019	1.948	1.990	2.027		2.006	2.07	2.00	2.03	2.175	2.181	2.171	2.171	2.121	2.164	2.034	2.078	2.052		
USA	1.862	1.819	1.784		1.818	1.826	1.845	1.809		1.817	1.91	1.81	1.79	1.804	2.006	1.832	1.803	1.778	1.895	1.819	1.768	1.808		

Source: Authors' computations.

Table S1.5. Correlations between the Percentage Variation of both Upstreamness and Downstreamness by Country Income Group for the Period 1995–2015

	dep variable:			
	$\Delta U\%$	$\Delta U\%$	$\Delta D\%$	$\Delta D\%$
ln GDP pc	0.021** (0.010)		0.022** (0.011)	
high_income=1		0.060*** (0.019)		0.067*** (0.022)
cons	-0.161* (0.093)	-0.014 (0.015)	-0.196** (0.099)	-0.038** (0.018)
R2	0.05	0.04	0.04	0.04
N	170	188	170	188

Source: Authors' computations.
 Note: Robust standard errors in parentheses: ***p < 0.01, **p < 0.05, *p < 0.1.

Table S1.6. Upstreamness by ISIC rev. 4 Sectors: A Comparison between OECD TiVA and WIOD for Overlapping Countries and Periods

sectors	2000			2001-2006			2007-2014			2000-2014		
	TiVA	WIOD		TiVA	WIOD		TiVA	WIOD		TiVA	WIOD	
1	2.184	2.211	Agriculture, hunting, forestry	2.232	2.272		2.458	2.603		2.408	2.498	
2	1.801	2.133	Fishing and aquaculture	1.854	2.158		2.255	2.408		2.153	2.333	
3to5	3.239	3.410	Mining and quarrying, energy producing products; Mining and quarrying, non-energy producing products; Mining support service activities	3.290	3.432		3.552	3.795		3.525	3.701	
6	1.519	1.629	Food products, beverages and tobacco	1.552	1.664		1.752	1.910		1.718	1.820	
7	1.862	1.973	Textiles, textile products, leather and footwear	1.853	1.978		2.177	2.323		2.160	2.198	
8	2.656	2.714	Wood and products of wood and cork	2.693	2.775		2.847	3.121		2.772	2.992	
9	2.907	3.054	Paper products and printing	2.894	3.044		3.019	3.200		2.977	3.134	
10	2.459	2.569	Coke and refined petroleum products	2.461	2.582		2.744	2.922		2.693	2.816	
11	2.970	3.030	Chemical and chemical products	2.994	3.050		3.296	3.329		3.211	3.232	
12	1.594	2.082	Pharmaceuticals, medicinal chemical and botanical products	1.588	2.063		1.662	2.092		1.659	2.082	
13	2.689	2.858	Rubber and plastics products	2.704	2.886		2.895	3.103		2.847	3.018	
14	2.498	2.645	Other non-metallic mineral products	2.586	2.646		2.635	2.740		2.601	2.708	
15	3.435	3.497	Basic metals	3.499	3.573		3.543	3.703		3.510	3.658	
16	2.553	2.631	Fabricated metal products	2.582	2.670		2.686	2.833		2.639	2.769	
17	2.040	2.214	Computer, electronic and optical equipment	2.046	2.285		2.216	2.553		2.194	2.449	
18	2.291	2.243	Electrical equipment	2.272	2.284		2.270	2.540		2.307	2.447	
19	1.921	1.934	Machinery and equipment, nec	1.939	1.998		2.027	2.211		2.022	2.136	
20	1.873	1.872	Motor vehicles, trailers and semi-trailers	1.903	1.887		1.947	2.018		1.917	1.968	
21	1.822	1.848	Other transport equipment	1.815	1.887		1.807	1.940		1.821	1.921	
22	1.901	1.851	Manufacturing nec; repair and installation of machinery and equipment	1.899	1.842		1.965	2.004		1.943	1.942	
23	2.412	2.469	Electricity, gas, steam and air conditioning supply	2.418	2.539		2.661	2.783		2.582	2.691	
24	2.026	2.459	Water supply; sewerage, waste management and remediation activities	2.133	2.491		2.265	2.675		2.244	2.607	
25	1.305	1.315	Construction	1.361	1.359		1.368	1.384		1.341	1.373	

Table S1.6. Continued

sectors	2000		2001-2006		2007-2014		2000-2014	
	TIVA	WIOD	TIVA	WIOD	TIVA	WIOD	TIVA	WIOD
26	2.042	1.822	2.041	1.832	2.157	1.954	2.119	1.909
27	2.232	2.348	2.305	2.394	2.470	2.544	2.409	2.490
28	2.639	2.812	2.689	2.899	2.895	3.037	2.842	2.985
29	2.096	1.988	2.131	2.026	2.218	2.208	2.184	2.136
30	2.604	2.910	2.685	2.967	2.822	3.077	2.768	3.038
31	2.838	2.832	2.838	2.855	2.886	2.937	2.873	2.903
32	1.494	1.492	1.465	1.480	1.503	1.563	1.515	1.533
33	2.020	2.112	2.002	2.065	2.018	2.091	2.001	2.083
34	2.172	2.182	2.135	2.147	2.051	2.122	2.073	2.133
35	2.087	2.095	2.103	2.119	2.113	2.162	2.124	2.146
36	2.211	2.325	2.228	2.336	2.393	2.476	2.348	2.424
37	1.417	1.420	1.440	1.454	1.457	1.474	1.460	1.466
38	2.473	2.594	2.488	2.605	2.568	2.683	2.540	2.655
39	2.653	2.667	2.666	2.654	2.768	2.693	2.724	2.679
40	1.126	1.177	1.126	1.167	1.147	1.172	1.143	1.171
41	1.111	1.162	1.119	1.175	1.116	1.181	1.115	1.179
42	1.060	1.073	1.061	1.078	1.062	1.082	1.061	1.080
43to44	1.490	1.506	1.484	1.513	1.465	1.587	1.474	1.560
45	1.000	1.407	1.000	1.133	1.000	1.123	1.000	1.139
Total	1.879	1.902	1.888	1.916	1.983	2.033	1.956	1.991

Source: Authors' computations.

Note: Sectors' positioning measures are weighted according to their relative gross output.

Table S1.7. Downstreamness by ISIC rev. 4 Sectors: A Comparison between WIOD and OECD TIVA for Overlapping Countries and Periods

Sectors	2000			2001-2006			2007-2014			2000-2014		
	TIVA	WIOD		TIVA	WIOD		TIVA	WIOD		TIVA	WIOD	
1 Agriculture, hunting, forestry	1.918	1.917		1.954	1.938		2.078	1.984		2.034	1.967	
2 Fishing and aquaculture	1.711	1.915		1.766	1.938		1.933	1.995		1.889	1.976	
3to5 Mining and quarrying, energy producing products; Mining and quarrying, non-energy producing products; Mining support service activities	1.836	1.797		1.858	1.751		1.833	1.653		1.879	1.683	
6 Food products, beverages and tobacco	2.419	2.444		2.454	2.486		2.602	2.675		2.550	2.607	
7 Textiles, textile products, leather and footwear	2.499	2.547		2.562	2.679		2.756	3.005		2.685	2.881	
8 Wood and products of wood and cork	2.466	2.371		2.466	2.441		2.555	2.598		2.497	2.531	
9 Paper products and printing	2.275	2.355		2.316	2.379		2.488	2.560		2.440	2.486	
10 Coke and refined petroleum products	2.314	2.497		2.286	2.473		2.466	2.631		2.423	2.585	
11 Chemical and chemical products	2.414	2.377		2.463	2.414		2.654	2.595		2.536	2.529	
12 Pharmaceuticals, medicinal chemical and botanical products	1.895	2.121		1.935	2.119		2.099	2.258		2.057	2.209	
13 Rubber and plastics products	2.367	2.500		2.422	2.589		2.588	2.806		2.496	2.719	
14 Other non-metallic mineral products	2.282	2.342		2.298	2.319		2.409	2.499		2.348	2.430	
15 Basic metals	2.555	2.592		2.682	2.747		2.874	3.015		2.752	2.939	
16 Fabricated metal products	2.319	2.381		2.416	2.478		2.625	2.793		2.569	2.676	
17 Computer, electronic and optical equipment	2.374	2.465		2.417	2.591		2.594	2.929		2.526	2.791	
18 Electrical equipment	2.459	2.535		2.513	2.651		2.844	3.134		2.726	2.966	
19 Machinery and equipment, nec	2.358	2.440		2.466	2.544		2.713	2.857		2.610	2.749	
20 Motor vehicles, trailers and semi-trailers	2.654	2.748		2.702	2.799		2.891	3.101		2.807	2.990	
21 Other transport equipment	2.394	2.423		2.432	2.507		2.681	2.887		2.600	2.770	
22 Manufacturing nec; repair and installation of machinery and equipment	2.187	2.235		2.241	2.272		2.415	2.412		2.352	2.355	
23 Electricity, gas, steam and air conditioning supply	1.958	2.019		1.994	2.050		2.261	2.252		2.156	2.178	
24 Water supply; sewerage, waste management and remediation activities	1.861	2.143		1.928	2.148		2.012	2.160		1.983	2.155	
25 Construction	2.210	2.215		2.268	2.303		2.503	2.626		2.440	2.513	

Table S1.7. Continued

Sectors	2000		2001-2006		2007-2014		2000-2014	
	TIVA	WIOD	TIVA	WIOD	TIVA	WIOD	TIVA	WIOD
26	1.685	1.664	1.715	1.682	1.774	1.737	1.752	1.716
27	1.820	1.918	1.877	1.976	2.007	2.123	1.951	2.068
28	2.301	2.361	2.357	2.404	2.466	2.575	2.431	2.515
29	2.214	2.177	2.239	2.196	2.356	2.390	2.254	2.312
30	1.938	1.969	1.985	2.022	2.054	2.096	2.033	2.069
31	1.691	1.782	1.770	1.814	1.905	1.931	1.848	1.882
32	1.920	1.969	1.947	1.967	2.032	2.064	1.999	2.028
33	2.028	2.017	1.950	1.927	1.928	1.875	1.922	1.901
34	1.939	1.913	1.913	1.906	1.960	1.930	1.921	1.921
35	1.782	1.840	1.716	1.757	1.886	1.871	1.825	1.836
36	1.785	1.870	1.755	1.831	1.824	1.886	1.772	1.867
37	1.413	1.411	1.455	1.440	1.452	1.401	1.446	1.414
38	1.789	1.724	1.788	1.752	1.916	1.840	1.882	1.808
39	1.710	1.611	1.689	1.588	1.730	1.702	1.691	1.662
40	1.641	1.639	1.672	1.652	1.713	1.700	1.686	1.682
41	1.403	1.440	1.412	1.442	1.448	1.544	1.430	1.510
42	1.660	1.717	1.667	1.722	1.722	1.800	1.707	1.773
43to44	1.763	1.742	1.784	1.768	1.799	1.845	1.767	1.816
45	1.000	1.160	1.000	1.159	1.000	1.126	1.000	1.138
Total	1.895	1.912	1.911	1.932	2.021	2.066	1.981	2.017

Source: Authors' computations.

Note: Sectors' positioning measures are weighted according to their relative gross output.

Table S1.8. Upstreamness by ISIC rev. 4 Sectors: A Comparison between ADB and WIOD for Overlapping Countries and Periods

sectors	2000		2007-2014		2000-2014	
	ADB	WIOD	ADB	WIOD	ADB	WIOD
1 Agriculture, hunting, forestry, and fishing	2.192	2.205	2.610	2.589	2.626	2.486
2 Mining and quarrying	3.343	3.410	3.704	3.795	3.765	3.701
3 Food, beverages, and tobacco	1.635	1.629	1.918	1.910	1.965	1.820
4ro5 Textiles and textile products; Leather, leather products, and footwear	1.971	1.973	2.320	2.323	2.360	2.198
6 Wood and products of wood and cork	2.699	2.714	3.102	3.121	3.113	2.992
7 Pulp, paper, paper products, printing, and publishing	2.642	2.634	2.712	2.650	2.731	2.618
8 Coke, refined petroleum, and nuclear fuel	2.539	2.569	2.847	2.922	2.907	2.816
9 Chemicals and chemical products	2.773	2.736	2.992	2.927	3.003	2.854
10 Rubber and plastics	2.849	2.858	3.085	3.103	3.155	3.018
11 Other nonmetallic minerals	2.644	2.645	2.738	2.740	2.719	2.708
12 Basic metals and fabricated metal	3.018	2.974	3.319	3.262	3.290	3.189
13 Machinery, nec	1.964	1.974	2.206	2.225	2.196	2.153
14 Electrical and optical equipment	2.211	2.222	2.531	2.549	2.561	2.449
15 Transport equipment	1.856	1.865	1.975	1.994	1.966	1.954
16 Manufacturing, nec; recycling	1.734	1.748	1.885	1.900	1.922	1.838
17 Electricity, gas, and water supply	2.420	2.439	2.702	2.720	2.729	2.639
18 Construction	1.314	1.315	1.383	1.384	1.371	1.373
19 Sale, maintenance, and repair of motor vehicles and motorcycles; retail sale of fuel	1.721	1.728	1.804	1.811	1.800	1.773
20 Wholesale trade and commission trade, except of motor vehicles and motorcycles	2.121	2.128	2.276	2.289	2.306	2.231
21 Retail trade, except of motor vehicles and motorcycles; repair of household goods	1.483	1.486	1.544	1.550	1.583	1.536
22 Hotels and restaurants	1.494	1.492	1.565	1.563	1.594	1.533
23 Inland transport	2.333	2.348	2.523	2.544	2.544	2.490
24 Water transport	2.786	2.812	3.021	3.037	3.071	2.985
25 Air transport	1.988	1.988	2.204	2.208	2.215	2.136
26 Other supporting and auxiliary transport activities; activities of travel agencies	2.905	2.910	3.067	3.077	3.111	3.038
27 Post and telecommunications	2.289	2.294	2.253	2.246	2.271	2.254
28 Financial intermediation	2.327	2.325	2.464	2.476	2.508	2.424
29 Real estate activities	1.423	1.420	1.476	1.474	1.498	1.466
30 Renting of M&Eq and other business activities	2.556	2.528	2.613	2.578	2.620	2.559
31 Public administration and defense; compulsory social security	1.176	1.177	1.169	1.172	1.158	1.171
32 Education	1.162	1.162	1.183	1.181	1.198	1.179
33 Health and social work	1.074	1.073	1.082	1.082	1.082	1.080
34 Other community, social, and personal services	1.748	1.629	1.840	1.749	1.856	1.712
35 Private households with employed persons	1.406	1.407	1.122	1.123	1.148	1.139
Total	1.902	1.902	2.033	2.033	2.046	1.991

Source: Authors' computations.

Note: Sectors' positioning measures are weighted according to their relative gross output.

Table S1.9. Downstreamness by ISIC rev. 4 Sectors: A Comparison between ADB and WIOD for Overlapping Countries and Periods

sectors	2000		2007-2014		2000-2014	
	ADB	WIOD	ADB	WIOD	ADB	WIOD
1 Agriculture, hunting, forestry, and fishing	1.907	1.916	1.991	1.985	1.953	1.968
2 Mining and quarrying	1.796	1.797	1.655	1.653	1.733	1.683
3 Food, beverages, and tobacco	2.434	2.444	2.664	2.675	2.655	2.607
4ro5 Textiles and textile products; Leather, leather products, and footwear	2.542	2.547	2.996	3.005	2.985	2.881
6 Wood and products of wood and cork	2.408	2.371	2.614	2.598	2.624	2.531
7 Pulp, paper, paper products, printing, and publishing	2.191	2.137	2.179	2.061	2.141	2.060
8 Coke, refined petroleum, and nuclear fuel	2.453	2.497	2.602	2.631	2.653	2.585
9 Chemicals and chemical products	2.323	2.275	2.518	2.450	2.493	2.392
10 Rubber and plastics	2.490	2.500	2.768	2.806	2.758	2.719
11 Other nonmetallic minerals	2.338	2.342	2.492	2.499	2.490	2.430
12 Basic metals and fabricated metal	2.458	2.407	2.897	2.837	2.883	2.722
13 Machinery, nec	2.403	2.411	2.791	2.806	2.798	2.702
14 Electrical and optical equipment	2.497	2.484	2.995	2.999	2.976	2.848
15 Transport equipment	2.663	2.680	3.024	3.045	3.018	2.936
16 Manufacturing, nec; recycling	2.250	2.254	2.444	2.453	2.405	2.388
17 Electricity, gas, and water supply	2.008	2.006	2.240	2.228	2.226	2.159
18 Construction	2.218	2.215	2.625	2.626	2.646	2.513
19 Sale, maintenance, and repair of motor vehicles and motorcycles; retail sale of fuel	1.741	1.746	1.823	1.831	1.796	1.802
20 Wholesale trade and commission trade, except of motor vehicles and motorcycles	1.712	1.710	1.762	1.765	1.751	1.749
21 Retail trade, except of motor vehicles and motorcycles; repair of household goods	1.605	1.606	1.692	1.693	1.689	1.668
22 Hotels and restaurants	1.966	1.969	2.062	2.064	2.058	2.028
23 Inland transport	1.913	1.918	2.117	2.123	2.106	2.068
24 Water transport	2.376	2.361	2.573	2.575	2.582	2.515
25 Air transport	2.185	2.177	2.400	2.390	2.354	2.312
26 Other supporting and auxiliary transport activities; activities of travel agencies	1.972	1.969	2.096	2.096	2.113	2.069
27 Post and telecommunications	1.883	1.907	1.930	1.930	1.951	1.920
28 Financial intermediation	1.846	1.870	1.832	1.886	1.820	1.867
29 Real estate activities	1.415	1.411	1.406	1.401	1.422	1.414
30 Renting of M&Eq and other business activities	1.739	1.765	1.839	1.830	1.845	1.801
31 Public administration and defense; compulsory social security	1.640	1.639	1.704	1.700	1.697	1.682
32 Education	1.441	1.440	1.547	1.544	1.608	1.510
33 Health and social work	1.730	1.717	1.824	1.800	1.865	1.773
34 Other community, social, and personal services	1.833	1.775	1.905	1.867	1.910	1.841
35 Private households with employed persons	1.159	1.160	1.126	1.126	1.106	1.138
Total	1.914	1.912	2.069	2.066	2.074	2.017

Source: Authors' computations.

Note: Sectors' positioning measures are weighted according to their relative gross output.

Table S1.10. Upstreamness by ISIC rev. 4 Sectors: A Comparison between ADB and OECD TiVA for Overlapping Countries and Periods

sectors	2000		2007-2014		2015		2016-2020		2000-2020	
	ADB	TiVA	ADB	TiVA	ADB	TiVA	ADB	TiVA	ADB	TiVA
1	2.192	2.156	2.610	2.444	2.775	2.676	2.646	2.446	2.626	2.390
2	3.343	3.239	3.704	3.552	3.995	3.715	3.852	3.689	3.765	3.525
3	1.635	1.519	1.918	1.752	2.078	1.978	2.037	1.819	1.965	1.718
4	2.011	2.363	2.363		2.587		2.456		2.398	
5	1.712	1.862	2.081	2.177	2.342	2.367	2.243	2.448	2.149	2.160
6	2.699	2.656	3.102	2.847	3.268	2.906	3.138	2.795	3.113	2.772
7,26,30	2.591	2.470	2.658	2.547	2.687	2.569	2.685	2.532	2.669	2.520
8	2.539	2.459	2.847	2.744	3.027	2.841	3.002	2.844	2.907	2.693
9	2.773	2.467	2.992	2.703	3.058	2.802	3.027	2.729	3.003	2.641
10	2.849	2.689	3.085	2.895	3.185	3.029	3.266	2.962	3.155	2.847
11	2.644	2.498	2.738	2.635	2.774	2.663	2.695	2.604	2.719	2.601
12	3.018	2.940	3.319	3.127	3.385	3.172	3.258	3.097	3.290	3.071
13,16	1.887	1.912	2.116	2.004	2.207	2.098	2.129	2.059	2.119	1.991
14	2.211	2.111	2.531	2.234	2.694	2.385	2.614	2.360	2.561	2.230
15	1.856	1.860	1.975	1.908	2.015	1.963	1.955	1.894	1.966	1.892
17,34	1.977	1.888	2.138	2.009	2.192	2.101	2.195	2.045	2.158	1.982
18	1.314	1.305	1.383	1.368	1.362	1.339	1.362	1.312	1.371	1.341
19	1.721		1.804		1.791		1.806		1.800	

Table S1.10. Continued

sectors	2000			2007-2014			2015			2016-2020			2000-2020		
	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA	ADB	TIVA	TIVA
20 Wholesale trade and commission trade, except of motor vehicles and motorcycles	2.121		2.276		2.384		2.346		2.306				2.306		
21 Retail trade, except of motor vehicles and motorcycles; repair of household goods	1.483	2.042	1.544	2.157	1.588	2.206	1.644	2.156	1.583	2.119			1.583	2.119	
22 Hotels and restaurants	1.494	1.494	1.565	1.503	1.624	1.601	1.638	1.569	1.594	1.515			1.594	1.515	
23 Inland transport	2.333	2.232	2.523	2.470	2.625	2.517	2.576	2.471	2.544	2.409			2.544	2.409	
24 Water transport	2.786	2.639	3.021	2.895	3.188	3.045	3.157	2.934	3.071	2.842			3.071	2.842	
25 Air transport	1.988	2.096	2.204	2.218	2.239	2.264	2.247	2.219	2.215	2.184			2.215	2.184	
27 Post and telecommunications	2.289	2.324	2.253	2.217	2.277	2.222	2.293	2.209	2.271	2.241			2.271	2.241	
28 Financial intermediation	2.327	2.211	2.464	2.393	2.571	2.499	2.564	2.400	2.508	2.348			2.508	2.348	
29 Real estate activities	1.423	1.417	1.476	1.457	1.497	1.492	1.534	1.501	1.498	1.460			1.498	1.460	
31 Public administration and defense; compulsory social security	1.176	1.126	1.169	1.147	1.180	1.161	1.137	1.154	1.158	1.143			1.158	1.143	
32 Education	1.162	1.111	1.183	1.116	1.191	1.123	1.221	1.115	1.198	1.115			1.198	1.115	
33 Health and social work	1.074	1.060	1.082	1.062	1.078	1.062	1.083	1.059	1.082	1.061			1.082	1.061	
35 Private households with employed persons	1.406	1.000	1.122	1.000	1.133	1.000	1.164	1.000	1.148	1.000			1.148	1.000	
Total	1.902	1.879	2.033	1.983	2.096	2.041	2.067	1.996	2.046	1.956			2.046	1.956	

Source: Authors' computations.

Note: Sectors' positioning measures are weighted according to their relative gross output.